

Convex-Treelike Partial Cubes: Peripheral Recursion and Forbidden Minors

Olivier Bousquet

Abstract

A tree-like partial cube is built from one vertex by peripheral expansions: each step duplicates an isometric subgraph of the graph already constructed. The duplicated subgraph need not itself be tree-like, so this class is not closed under convex subgraphs. Polat’s convex-treelike partial cubes require every finite nontrivial convex subgraph to have a periphery. We characterize the finite members by the stronger peripheral recursion in which both the contraction base and the duplicated subgraph admit the same construction.

We prove that this class is closed under partial-cube minors and is the largest such subclass of the tree-like partial cubes. Its description as the largest convex-subgraph-closed subclass follows from Polat’s characterization. Here partial-cube minors use coordinate projection and restriction to convex subgraphs. We then characterize membership and minimal forbidden minors for one-vertex amalgams. The relevant pointed factors are those whose peripheral decompositions can be chosen to remove sides avoiding the identified vertex. Minimal failures of this pointed property generate exactly the one-vertex-decomposable obstructions.

The forbidden minors with nonample vertex sets are precisely the cubes with two antipodal vertices deleted. For the remaining, ample obstructions, we give exact coordinate recognition criteria and express the effect of projection through minimal sets of coordinates that repair a failed section containment. We also give explicit obstructions and computer-assisted small-order classifications. The recognition criteria may inspect all coordinate states; a simpler intrinsic generator for the full forbidden family remains open.

1 Introduction

Expansion is one of the fundamental ways to construct a partial cube. Contracting a Djoković–Winkler class deletes one intrinsic hypercube coordinate; reversing the contraction splits a graph along two isometric covering subgraphs and adds a new matching. Classical restrictions on the overlap lead to median, almost-median, and semi-median graphs [IK98, BIK03a]. A different restriction is *peripherality*: one member of the cover is the entire base graph. Brešar, Imrich, and Klavžar used sequences of such expansions to define the tree-like partial cubes [BIK03b].

There is a quantifier hidden in that definition. A sequential construction records only one chain of contraction bases. The second member of a peripheral cover—the site duplicated by the new coordinate—need not have a compatible peripheral construction. This distinction matters structurally. Tree-like partial cubes are closed under gated subgraphs, but not under convex subgraphs [Pol12, Gol14]; in fact every finite partial cube occurs as a convex semicube of some tree-like partial cube. Hence the standard tree-like class cannot be characterized by forbidden partial-cube minors.

Polat introduced *convex-treelike partial cubes*: every finite nontrivial convex subgraph must have a periphery [Pol12, Definition 5.5]. He also proved that this is equivalent to requiring every finite convex subgraph to be tree-like [Pol12, Proposition 5.6]. Throughout this paper our graphs are finite.

We study this established class through a recursive construction and its forbidden partial-cube minors. Starting with K_1 , allow a peripheral expansion only when both its contraction base and its duplicated site have already been constructed by the same rule. We denote the resulting class by $\mathcal{R}_{\text{per}}$. We use *cover-recursive peripheral* to describe this construction; [theorem 4.1](#) identifies its members with the finite convex-treelike partial cubes. The requirement is motivated by the established distinction between sequential and cover-recursive expansion classes: for example, proper semi-median graphs require both cover members recursively, whereas the isometric-expansion class uses only one expansion sequence [\[BIK⁺02\]](#).

Main results. Our first theorem shows that the recursive condition supplies exactly the heredity missing from the standard tree-like class.

Theorem 1.1 (Hereditary characterization; informal form). *For a finite partial cube G , the following are equivalent:*

- (i) $G \in \mathcal{R}_{\text{per}}$;
- (ii) every nontrivial partial-cube minor of G has a peripheral coordinate;
- (iii) every partial-cube minor of G is tree-like;
- (iv) every nontrivial convex subgraph of G has a peripheral coordinate; and
- (v) every convex subgraph of G is tree-like.

The equivalence of (iv) and (v) is Polat’s characterization. We first identify his class with the recursive construction: a convex subgraph that meets both layers of a peripheral expansion inherits that expansion. We then commute coordinate contraction past one peripheral step. These two arguments give [theorem 4.1](#) and [lemma 5.1](#); the pc-minor tests are consequences. Expansion trees record the recursive construction.

Our second structural result concerns gluing at a vertex. A peripheral decomposition of a pointed graph is *root-avoiding* if each removed peripheral side omits the root and this condition continues on the contraction base. The duplicated site needs only an ordinary decomposition. For two members of $\mathcal{R}_{\text{per}}$, their one-vertex amalgam belongs to $\mathcal{R}_{\text{per}}$ if and only if at least one factor has a root-avoiding decomposition at the identified vertex. Moreover, the amalgam is a minimal forbidden pc-minor if and only if both factors are minimal failures of that pointed property under root-preserving minors; see [theorem 10.5](#). This gives a construction of all one-vertex-decomposable obstructions. It also provides an explicit failure of closure under gated amalgamation.

The class contains every median graph and every daisy cube, and all its members are tree-like, ample, and corner-peelable. Here an ample class is a vertex family whose shattered coordinate sets are exactly the coordinate sets of its contained cubes. A corner peeling successively deletes vertices lying in a unique maximal cube; the equivalent isometric-ordering convention is fixed in [section 2](#). The inclusions and separating examples are collected in [sections 7 to 9](#).

The forbidden family has both a structural and an algorithmic description. Its nonample members are exactly $Q_n^- = Q_n - \{x, \bar{x}\}$, $n \geq 3$. Every other forbidden minor is ample. The sixteen-vertex pentagonal class $\mathcal{P}_5$ is an explicit example; its global vertex-minimality among these additional obstructions is computer-assisted. The one-vertex construction gives a twenty-one-vertex example from two eleven-vertex pointed factors.

For recognition, assign each coordinate one of four instructions: retain, project, fix to zero, or fix to one. A nontrivial irredundant partial cube is forbidden exactly when it has no peripheral

coordinate and every proper nontrivial resulting minor has one. This follows from the hereditary characterization. The later two-layer formulation records witnesses to failed section containments and the minimal coordinate sets whose projection repairs them. These are exact tests, potentially over exponentially many coordinate states. They do not yet give a succinct intrinsic generator comparable to a decomposition into a few elementary constructions.

Results at a glance. The paper separates equivalent characterizations from consequences and from finite computations. The following table records what each part contributes.

Viewpoint	What it provides	Location
Expansions	Binary peripheral expansion trees, and the precise distinction from a sequential tree-like construction	Section 3
Convex subgraphs	Identification with Polat’s class and its convex-subgraph test	Section 4
Pc-minors	Hereditiy and the test that every nontrivial pc-minor exposes a peripheral class	Section 5
Forbidden basis	A coordinate recognition criterion and the nonample family; later, two-layer recognition and repair antichains	Sections 6 and 11
Structure and examples	Class containments, closure properties, explicit obstructions, and the exact one-vertex-amalgam theorem	Sections 7 to 10
Finite verification	Finite classifications, enumeration domains, and verification methods	Section 12

Provenance and computation. The expansion, tree-like, pc-minor, ample, and corner-peeling results quoted as background are explicitly attributed to their sources. The class definition and the equivalence of the two convex-subgraph tests are due to Polat; the largest convex-hereditary subclass statement is a consequence of that equivalence. We give proofs of the recursive and pc-minor characterizations, the rooted-wedge theorem, the obstruction minimality arguments, and the small-order lemma. Polat’s pointed example already contains the factor used in our twenty-one-vertex obstruction; we explain the relation and extend his construction in [section 10.2](#). Finite uniqueness and small-order assertions are computer-assisted. The enumeration is exhaustive up to cube isomorphism and is documented in [section 12](#). Symbolic proofs do not rely on the enumeration unless a statement is explicitly labelled computer-assisted.

Reading guide. [Section 2](#) fixes the partial-cube, minor, expansion, and set-system language. [Section 3](#) gives the expansion-tree characterization and separates it from the usual sequential tree-like construction. [Section 4](#) identifies the recursion with Polat’s class, and [section 5](#) adds contraction closure and the pc-minor characterization. [Section 6](#) gives coordinate recognition and identifies the nonample obstructions. [Section 7](#) positions the class among established classes, and [section 8](#) collects its closure and inherited properties. [Section 9](#) gives explicit ample obstructions and the global small-order result. [Section 10](#) treats gated and one-vertex amalgams. The main structural arguments can be read without the later repair machinery: [section 11](#) develops that coordinate test after the examples and wedge theorem. [Section 12](#) explains the finite enumerations, and [section 13](#) states the remaining generator problem.

2 Preliminaries

All graphs and coordinate sets are finite. Graphs are simple and connected unless stated otherwise. For a graph G , let d_G denote its shortest path metric and

$$I_G(u, v) = \{w : d_G(u, w) + d_G(w, v) = d_G(u, v)\}$$

its interval. A vertex set S is *convex* if $I_G(u, v) \subseteq S$ for all $u, v \in S$; it is *gated* if every vertex x has a unique $g \in S$ such that

$$d_G(x, s) = d_G(x, g) + d_G(g, s) \quad (s \in S).$$

Gated subgraphs are convex, and convex subgraphs are isometric.

2.1 Coordinates, sections, and partial-cube minors

The X -dimensional hypercube Q_X has vertex set $\{0, 1\}^X$ and Hamming distance. A *partial cube* is a graph admitting an isometric embedding into a hypercube. We always suppress constant coordinates, so the embedding is irredundant.

For edges xy and uv , the Djoković–Winkler relation is

$$xy \Theta uv \iff d(x, u) + d(y, v) \neq d(x, v) + d(y, u).$$

In a partial cube this is an equivalence relation. Its classes are exactly the edge sets that change one coordinate in an irredundant embedding; their number is the *isometric dimension* $\text{idim}(G)$.

Fix an embedding $V(G) \subseteq \{0, 1\}^X$. For $x \in X$ and $\varepsilon \in \{0, 1\}$, the two semicubes and their boundaries are

$$W_x^\varepsilon = \{v : v_x = \varepsilon\}, \quad U_x^\varepsilon = \{v \in W_x^\varepsilon : v \text{ has a neighbor in } W_x^{1-\varepsilon}\}.$$

The semicubes are complementary and convex. Identifying a binary vector with its support gives a family $H \subseteq 2^X$. After deleting coordinate x , its two sections and its projection are

$$H_x^\varepsilon = \{h \setminus \{x\} : h \in H, \mathbf{1}_{x \in h} = \varepsilon\}, \quad \pi_x H = H_x^0 \cup H_x^1.$$

Restricting G to W_x^ε gives the graph of H_x^ε , while contracting the entire x -class gives the graph of $\pi_x H$.

Definition 2.1 (Peripheral coordinate). A Θ -class x is *peripheral* if $U_x^\varepsilon = W_x^\varepsilon$ for at least one $\varepsilon \in \{0, 1\}$. Equivalently, its projected sections are nested:

$$H_x^\varepsilon \subseteq H_x^{1-\varepsilon}$$

for one orientation. The smaller semicube is a *periphery*. A nontrivial partial cube with no peripheral class is *periphery-free*.

Definition 2.2 (Partial-cube minor). A *restriction* of G is a nonempty convex subgraph. The *contraction* G/E of a Θ -class E simultaneously identifies the endpoints of all edges of E and takes the simple quotient. A *partial-cube minor*, or *pc-minor*, is obtained by a finite sequence of restrictions and Θ -class contractions. It is *proper* if it is not isomorphic to G .

It suffices to use elementary restrictions to one semicube and contractions of one class [KM20, Section 2.1]. Every convex subgraph of a finite partial cube is an intersection of semicubes. In set-family language, the elementary operations are H_x^ε and $\pi_x H$, respectively.

2.2 Shattering and neighboring graph classes

For $H \subseteq 2^X$ and $T \subseteq X$, the trace is $H|_T = \{h \cap T : h \in H\}$. The family H *shatters* T when $H|_T = 2^T$, and *strongly shatters* T when it contains a T -dimensional cube with all other coordinates fixed. Write

$$\text{str}(H) = \{T : H \text{ shatters } T\}, \quad \text{sstr}(H) = \{T : H \text{ strongly shatters } T\}.$$

The VC dimension is the largest size of a shattered set. The sandwich inequalities are

$$|\text{sstr}(H)| \leq |H| \leq |\text{str}(H)|.$$

A family is *ample*, or *lopsided*, when equality holds, equivalently when $\text{sstr}(H) = \text{str}(H)$ [BCDK06]. A VC-dimension- r family on n coordinates is *maximum* when it has $\sum_{i=0}^r \binom{n}{i}$ members. Every maximum family is ample.

The *crossing graph* $G^\sharp$ has the Θ -classes as vertices; two are adjacent when all four intersections of their semicubes are nonempty, equivalently when their coordinate pair is shattered. For an ample partial cube this agrees with the graph recording pairs realized by a square.

We shall compare $\mathcal{R}_{\text{per}}$ with the following established classes.

- A partial cube is *median* if, for every three vertices u, v, w , the intersection $I_G(u, v) \cap I_G(v, w) \cap I_G(w, u)$ has exactly one vertex.
- It is *almost-median*, respectively *semi-median*, if every boundary graph $G[U_x^\varepsilon]$ is isometric, respectively connected.
- It is *proper semi-median* if it is semi-median and admits a connected expansion cover (one with connected overlap) whose two members recursively have the same property, beginning with K_1 [BIK⁺02, Section 3, class (10)].
- It is a *daisy cube* if, after choosing a root and orientations of the coordinates, its vertex family is a down-set in 2^X .

For an ample family, call an ordering $v_1, \dots, v_m$ *prefix-isometric* when every induced prefix $\{v_1, \dots, v_i\}$ is isometric. Chalopin, Chepoi, Moran, and Warmuth prove that these are exactly the corner-peeling orderings of an ample family [CCMW22, Proposition 4.2]. We use this equivalence as the definition-independent meaning of *corner-peelable*. Following that source, an ample partial cube admitting such an ordering is also called *dismantlable ample*; this is unrelated to closed-neighborhood dismantlability for general graphs.

Reversing a prefix-isometric ordering gives the order in which corners are deleted. Statements about a corner peeling *ending* at a specified vertex use this deletion order; the corresponding prefix ordering begins at that vertex.

Example 2.3 (Three small coordinate models). The path P_4 , the cycle C_6 , and the cube Q_3 have the representations

G	H	$\text{idim}(G)$	$G^\sharp$
P_4	$\{\emptyset, \{1\}, \{1, 2\}, [3]\}$	3	$\bar{K}_3$
C_6	$2^{[3]} \setminus \{\emptyset, [3]\}$	3	K_3
Q_3	$2^{[3]}$	3	K_3 .

The 6-cycle shatters every coordinate pair but contains no square. It is the smallest antipodally punctured cube Q_3^{--} and is periphery-free. The cube is recursively peripheral, yet contains this C_6 as an induced isometric subgraph.

2.3 Expansion and its quantifiers

Definition 2.4 (Expansion). An *expansion cover* of a connected graph B is a pair of isometric induced subgraphs B_0, B_1 such that

$$V(B) = V(B_0) \cup V(B_1), \quad E(B) = E(B_0) \cup E(B_1), \quad V(B_0) \cap V(B_1) \neq \emptyset.$$

The corresponding *expansion* takes disjoint copies of B_0, B_1 and joins the two copies of every vertex in $B_0 \cap B_1$. It is *peripheral* when one cover member is the whole base B .

The expansion is *connected* when its overlap $B_0 \cap B_1$ is connected, and *isometric* when that overlap is isometric in B . The latter condition is stronger than requiring the two cover members to be isometric, which is part of every expansion cover here.

Every finite partial cube is obtained from K_1 by expansions, reversing a complete sequence of Θ -class contractions [IK98, Theorem 3.1]. In a peripheral expansion, orient the new coordinate so that

$$V(G) = (V(B) \times \{0\}) \cup (V(S) \times \{1\}), \quad S \subseteq B, \quad (1)$$

where the duplicated site S is isometric in B .

There are two ways to generate a class from this operation.

Definition 2.5 (Sequential and cover-recursive peripheral constructions). A *sequential peripheral construction* is a chain

$$K_1 = G_0 \longrightarrow G_1 \longrightarrow \cdots \longrightarrow G_m = G$$

of peripheral expansions. The graphs admitting such a construction are the standard *tree-like partial cubes* [BIK03b].

The *cover-recursive peripheral class* $\mathcal{R}_{\text{per}}$ is the least class containing K_1 such that a graph in the form (1) belongs to $\mathcal{R}_{\text{per}}$ whenever both B and S belong to $\mathcal{R}_{\text{per}}$.

By theorem 4.1, $\mathcal{R}_{\text{per}}$ is precisely the finite convex-treelike class of [Pol12, Definition 5.5]. We retain the recursive notation to make the two-child construction explicit.

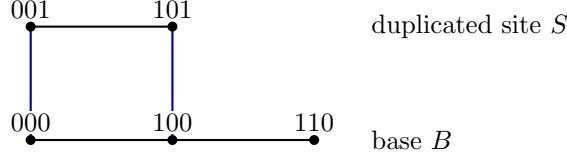
Equivalently, a nontrivial partial cube G belongs to $\mathcal{R}_{\text{per}}$ when *there exists* a peripheral coordinate e for which both the contraction G/e and the duplicated site belong to $\mathcal{R}_{\text{per}}$. Only one coordinate is required to witness a recursive step. The requirement is not universal over the peripheral coordinates, much less over all coordinates. The recursion is instead universal over the two children of each chosen step. Thus its witness is a rooted binary tree, whereas a tree-like witness is one distinguished path of contraction bases.

Class	Witness	Requirement at one peripheral step
tree-like partial cubes	one path of contraction bases	the contraction base continues the path; no recursive condition is imposed on the duplicated site
$\mathcal{R}_{\text{per}}$	a rooted binary tree	both the contraction base and the duplicated site recursively belong to $\mathcal{R}_{\text{per}}$

Example 2.6 (A square with an attached edge). Take the path $B = \{00, 10, 11\}$ and its edge $S = \{00, 10\}$. Duplicating S gives

$$G = \{000, 100, 110, 001, 101\}.$$

The last coordinate is peripheral: its one-section is S , contained in its zero-section B . Projecting that coordinate recovers B ; conditioning it to one recovers S . Both are paths and admit recursive peripheral decompositions, so $G \in \mathcal{R}_{\text{per}}$.



The two vertical edges form the new coordinate class. The two recursive children are the three-vertex path and the edge, rather than just the path.

Remark 2.7 (Why the two classes differ). Embed any finite partial cube H isometrically in Q_n , and expand Q_n peripherally along H . The result is tree-like, while restriction to the new semicube recovers H . Consequently every finite partial cube is a pc-minor of a tree-like partial cube. In particular, the tree-like class is not pc-minor closed and cannot equal $\mathcal{R}_{\text{per}}$.

3 Characterization by peripheral expansion trees

The definition of $\mathcal{R}_{\text{per}}$ is recursive rather than sequential. The following certificate makes the distinction visible and will be useful when we compare $\mathcal{R}_{\text{per}}$ with the tree-like partial cubes.

Definition 3.1 (Peripheral expansion tree). A *peripheral expansion tree* for a finite partial cube G is a finite rooted binary tree whose nodes are labelled by partial cubes. The root is labelled by G , every leaf by K_1 , and every internal node labelled by H has the following form for some peripheral coordinate e :

$$H = (B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B,$$

with its two children labelled by the contraction base $B = H/e$ and the duplicated site S , respectively.

Theorem 3.2 (Expansion-tree characterization). *A finite partial cube belongs to $\mathcal{R}_{\text{per}}$ if and only if it admits a peripheral expansion tree.*

Proof. Read a tree from its leaves toward its root. Each internal node is obtained from its two children by the closure rule in [definition 2.5](#), so its root belongs to $\mathcal{R}_{\text{per}}$. Conversely, unfold a recursive construction of a member of $\mathcal{R}_{\text{per}}$: at each nontrivial node attach its contraction base and duplicated site as the two children. Their isometric dimensions are smaller, so the unfolding is finite and all leaves are singletons. $\square$

A sequential peripheral construction retains only the successive base children of this tree. Thus it is one root-to-leaf path, with no condition on the off-path sites. The next example shows that the omitted branches are an actual obstruction, not merely extra bookkeeping.

Example 3.3 (A sequential construction with a bad site). Let

$$B = Q_3 - \{111\}, \quad S = \{x \in \{0, 1\}^3 : 1 \leq |x| \leq 2\} \cong C_6,$$

and peripherally expand B along the isometric site S :

$$T = (B \times \{0\}) \cup (S \times \{1\}).$$

The base B is a daisy cube and hence tree-like, so this is a sequential peripheral construction of T . The new coordinate is peripheral because $S \subseteq B$. For any old coordinate p , symmetry permits us to write the other two old coordinates first and the new coordinate last. The pattern 110 occurs only in the $p = 0$ section, whereas 001 occurs only in the $p = 1$ section. Thus neither old section

contains the other, and the new coordinate is the unique peripheral one. Its duplicated site $S \cong C_6$ is periphery-free, so the unique possible first branch cannot be completed to a peripheral expansion tree. Hence T is tree-like but does not belong to $\mathcal{R}_{\text{per}}$. The construction of a tree-like expansion of $Q_3 - \{111\}$ with a convex six-cycle already appears in [Pol12, Figure 2 and the discussion after Definition 5.5]; see also [Gol14, Figure 2].

4 Convex-treelike graphs and peripheral recursion

Polat's definition gives a hereditary starting point for the recursive construction. The equivalence of (ii) and (iii) below is [Pol12, Proposition 5.6]. We prove the connection with the expansion rule directly, without using contraction closure.

Theorem 4.1 (Convex-subgraph characterization). *For a finite partial cube G , the following are equivalent:*

- (i) $G \in \mathcal{R}_{\text{per}}$;
- (ii) every convex subgraph of G with at least two vertices has a peripheral Θ -class;
- (iii) every convex subgraph of G is tree-like.

Consequently, $\mathcal{R}_{\text{per}}$ is the finite convex-treelike class and is the largest convex-subgraph-closed subclass of the tree-like partial cubes.

Proof. First we check convex-subgraph closure of the recursive class by induction on isometric dimension. Write a nontrivial recursive member as $G = (B \times \{0\}) \cup (S \times \{1\})$, with $S \subseteq B$ and $B, S \in \mathcal{R}_{\text{per}}$, and let H be a nonempty convex subgraph of G . Its intersections with the two layers, projected to B and S , are convex subgraphs H_0, H_1 of those graphs. If only one is nonempty, induction applies to it.

If both are nonempty, then $H_1 \subseteq H_0$. Indeed, for $x \in H_1$ and $y \in H_0$, the edge from $(x, 1)$ to $(x, 0)$, followed by a geodesic in B to $(y, 0)$, is a geodesic of G . Convexity of H forces $(x, 0) \in H$. Induction puts H_0, H_1 in $\mathcal{R}_{\text{per}}$; their peripheral expansion is H . Thus every convex subgraph of a recursive member is recursive. Nontrivial recursive graphs have a periphery and are tree-like, proving (i) implies (ii) and (iii).

Conversely, assume (ii). A nontrivial G has a peripheral coordinate. Its two semicubes are convex and are isomorphic to the contraction base and duplicated site. Both inherit (ii), so induction on isometric dimension constructs both recursively and then constructs G . The singleton is the base case. Finally, (iii) implies (ii) because every nontrivial tree-like graph has a periphery. The largest-subclass assertion follows from (iii). $\square$

5 Characterization by partial-cube minors

Convex-subgraph closure is now available from [theorem 4.1](#). The additional step needed for pc-minors is closure under coordinate contraction. Polat's self-contractions are nonexpansive self-maps [Pol12, Section 3]; those results do not assert closure under the coordinate quotients considered here.

Lemma 5.1 (Pc-minor stability). *The class $\mathcal{R}_{\text{per}}$ is closed under restriction to a nonempty semicube and under contraction of an arbitrary Θ -class. Consequently it is closed under convex subgraphs and pc-minors.*

Proof. Only contractions remain to be checked. Induct on $d = \text{idim}(G)$, with K_1 as the base case, and write

$$G = (B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B, \quad B, S \in \mathcal{R}_{\text{per}},$$

using a witnessing peripheral coordinate e . Contracting e gives B . For a different coordinate f ,

$$\pi_f G = (\pi_f B \times \{0\}) \cup (\pi_f S \times \{1\}).$$

Coordinate contraction preserves isometric cube embeddings, so $\pi_f S$ is isometric in $\pi_f B$. Both graphs lie in $\mathcal{R}_{\text{per}}$ by induction, since B, S have isometric dimension at most $d-1$. If f is constant on S , its deletion simply leaves an isomorphic copy of S . The displayed peripheral expansion therefore belongs to $\mathcal{R}_{\text{per}}$. Combining this with convex-subgraph closure proves pc-minor closure. $\square$

Theorem 5.2 (Partial-cube-minor characterization). *For a finite partial cube G , the following are equivalent:*

- (i) $G \in \mathcal{R}_{\text{per}}$;
- (ii) every pc-minor of G with at least two vertices has a peripheral Θ -class;
- (iii) every pc-minor of G is tree-like;

Consequently, $\mathcal{R}_{\text{per}}$ is the largest pc-minor-closed subclass of the tree-like partial cubes.

Proof. If (i) holds, lemma 5.1 puts every pc-minor in $\mathcal{R}_{\text{per}}$, and theorem 4.1 gives (ii) and (iii). Either (ii) or (iii), applied just to the convex subgraphs of G , gives (i) by that same characterization. Finally, every pc-minor of a member of a pc-minor-closed subclass of the tree-like graphs is tree-like; (iii) therefore puts that subclass inside $\mathcal{R}_{\text{per}}$. $\square$

Convex heredity upgrades the existential choice in the expansion definition to a choice-independent one.

Corollary 5.3 (Every peripheral coordinate is admissible). *Let $G \in \mathcal{R}_{\text{per}}$, and let e be any peripheral coordinate of G . Then the contraction base and duplicated site at e both belong to $\mathcal{R}_{\text{per}}$. Consequently e may label the root of a peripheral expansion tree for G , and arbitrary peripheral choices may be made independently at all later nodes.*

Proof. The contraction base and duplicated site are isomorphic to the two convex semicubes of G , so theorem 4.1 puts both in $\mathcal{R}_{\text{per}}$. Apply theorem 3.2 to both children and attach their trees below the chosen root. Repeating the same argument at every node proves the final assertion. $\square$

Corollary 5.4 (Forbidden-minor form). *Let $\mathcal{P}$ be the family of finite nontrivial periphery-free partial cubes, and let $\mathcal{O}_{\text{per}}$ be its pc-minor-minimal members. Then*

$$\mathcal{R}_{\text{per}} = \mathcal{F}(\mathcal{P}) = \mathcal{F}(\mathcal{O}_{\text{per}}),$$

where $\mathcal{F}(\mathcal{X})$ denotes the partial cubes having no member of $\mathcal{X}$ as a pc-minor. Equivalently, $\mathcal{O}_{\text{per}} = \text{Obs}_{\text{pc}}(\mathcal{R}_{\text{per}})$.

Proof. By theorem 5.2, a partial cube lies outside $\mathcal{R}_{\text{per}}$ exactly when some pc-minor is nontrivial and periphery-free. Taking a minimal such minor gives a member of $\mathcal{O}_{\text{per}}$. Conversely, every proper pc-minor of a member of $\mathcal{O}_{\text{per}}$ has no periphery-free pc-minor and therefore belongs to $\mathcal{R}_{\text{per}}$, proving the final equality. $\square$

6 The forbidden partial-cube minors

The intrinsic characterization in [theorem 5.2](#) says that the forbidden minors are the pc-minor-minimal partial cubes without a peripheral coordinate. We first give a direct coordinate-state formulation involving only section containments and identify the nonample family. The more detailed two-layer and repair-antichain tests are postponed to [section 11](#); they are not needed for the structural arguments and examples that follow.

6.1 A direct coordinate-state characterization

Let $O \subseteq \{0, 1\}^X$ be irredundant. A *full minor state* is a map

$$\tau : X \longrightarrow \{\mathbf{K}, \mathbf{C}, 0, 1\},$$

whose four values mean retain, contract, restrict to zero, and restrict to one. Put

$$K_\tau = \tau^{-1}(\mathbf{K}), \quad C_\tau = \tau^{-1}(\mathbf{C}), \quad F_\tau = \tau^{-1}(\{0, 1\}),$$

and define the resulting coordinate family directly by

$$O[\tau] = \{x|_{K_\tau} : x \in O, x_q = \tau(q) \text{ for every } q \in F_\tau\}. \quad (2)$$

The state is *feasible* when $O[\tau] \neq \emptyset$. Let $\text{var}(O[\tau]) \subseteq K_\tau$ be the coordinates that are nonconstant on $O[\tau]$, and let $\mathbf{K}_X$ denote the all-retained state.

Theorem 6.1 (Coordinate-state description of the complete basis). *Let $O \subseteq \{0, 1\}^X$ be an irredundant finite partial cube with at least two vertices. Then $O \in \mathcal{O}_{\text{per}}$ if and only if the following two conditions hold:*

- (i) *for every $p \in X$ and $b \in \{0, 1\}$,*

$$O_p^{1-b} \not\subseteq O_p^b; \quad (3)$$

- (ii) *for every feasible state $\tau \neq \mathbf{K}_X$ with $|\text{var}(O[\tau])| \geq 2$, there are $p \in \text{var}(O[\tau])$ and $b \in \{0, 1\}$ such that*

$$O[\tau]_p^{1-b} \subseteq O[\tau]_p^b. \quad (4)$$

In words, the original family has no peripheral coordinate, while every proper nontrivial coordinate state exposes one.

Proof. Condition (3) is exactly the assertion that neither orientation of any coordinate of O is peripheral.

Every sequence of elementary pc-minor operations has the form (2): operations in distinct coordinates commute, a coordinate disappears after its first operation, restrictions contribute the fixed coordinates F_τ , and contractions contribute C_τ . Conversely, every feasible state is realized by first applying its restrictions and then its contractions. The restrictions can be ordered arbitrarily, because a preimage of any member of $O[\tau]$ keeps every prefix nonempty. Deleting the coordinates outside $\text{var}(O[\tau])$ only makes the representation irredundant. Thus the nonidentity feasible states are exactly the proper pc-minors, up to the deletion of constant coordinates.

Suppose first that $O \in \mathcal{O}_{\text{per}}$. It is periphery-free: a peripheral coordinate would have a contraction base and duplicated site among its proper minors, and their permitted expansion would put O in $\mathcal{R}_{\text{per}}$. This proves (i). Every proper pc-minor belongs to $\mathcal{R}_{\text{per}}$, hence every nontrivial one has a peripheral coordinate. This is (ii).

Conversely, assume (i)–(ii). The family O is outside $\mathcal{R}_{\text{per}}$ by [theorem 5.2](#). Let N be a proper pc-minor. Every nontrivial pc-minor of N is itself a proper pc-minor of O , so (ii) gives it a peripheral coordinate. Applying [theorem 5.2](#) to N yields $N \in \mathcal{R}_{\text{per}}$. Therefore all proper pc-minors of O lie in $\mathcal{R}_{\text{per}}$, and $O \in \mathcal{O}_{\text{per}}$. $\square$

Remark 6.2 (What the coordinate-state theorem provides). Theorem [6.1](#) recognizes the forbidden family from section containments. Its direct implementation may inspect all $4^{|X|}$ states. The repair-antichain formulation below evaluates the same conditions from the original family and explains how a failed containment can disappear under contraction. The hypothesis of at least two vertices is essential: on the empty coordinate set both displayed conditions are vacuous, whereas the singleton belongs to $\mathcal{R}_{\text{per}}$.

The minimality conditions also give every obstruction a canonical description as a member of the class with a few vertices removed. For a coordinate e of O and $b \in \{0, 1\}$, call

$$\Delta_e^b(O) = O_e^b \setminus O_e^{1-b}$$

the b -sided defect of e . The coordinate e is peripheral exactly when one of its two defects is empty.

Proposition 6.3 (Deletion normal form). *Let $O \in \mathcal{O}_{\text{per}}$, let e be a coordinate of O , let $b \in \{0, 1\}$, and put $D = \Delta_e^b(O)$. Then*

$$G = O \cup \{(x, 1-b) : x \in D\}$$

is a peripheral expansion of O/e along the site O_e^b , so $G \in \mathcal{R}_{\text{per}}$, and

$$O = G \setminus \{(x, 1-b) : x \in D\}.$$

Here G is a partial cube whose coordinate e is peripheral, and the deleted vertices lie in the $(1-b)$ -semicube of e and have their e -neighbours in G .

Proof. The $(1-b)$ -section of G is $O_e^{1-b} \cup D = O_e^0 \cup O_e^1 = O/e$, and its b -section is O_e^b , which is contained in O/e . Thus $G = ((O/e) \times \{1-b\}) \cup (O_e^b \times \{b\})$ in the coordinate order that puts e last. The site O_e^b and the base O/e are proper pc-minors of O , hence belong to $\mathcal{R}_{\text{per}}$; in particular the site is a partial cube, so G is a partial cube. The recursive definition gives $G \in \mathcal{R}_{\text{per}}$. The displayed equality is the definition of G . $\square$

Every obstruction is therefore obtained from a member of $\mathcal{R}_{\text{per}}$ by deleting the smaller defect of a suitable coordinate. For Q_n^{--} , for $\mathcal{P}_5$, and for the cyclic-arc classes of [theorem 9.4](#), one vertex suffices. One vertex does not always suffice: for the step-two classes of [theorem 9.7](#), every defect has at least three vertices. The converse direction is the difficulty. By [theorem 9.9](#), whether a deletion from a class member is an obstruction cannot be decided from the pc-minors of bounded depth.

6.2 The nonample family

We first dispose of the nonample case. For $n \geq 3$, write

$$Q_n^{--} = Q_n - \{0^n, 1^n\}, \quad Q_n^- = Q_n - \{0^n\}.$$

Proposition 6.4 (The nonample obstructions). *For every $n \geq 3$, the graph Q_n^{--} is a pc-minor-minimal graph outside $\mathcal{R}_{\text{per}}$. Consequently $\mathcal{O}_{\text{per}}$ is infinite and $\mathcal{R}_{\text{per}}$ has no finite forbidden-pc-minor characterization.*

Proof. First, Q_n^{--} is a partial cube. A cube geodesic between two retained vertices cannot pass through both deleted antipodes, because two vertices at distance n on a geodesic must be its endpoints. If it passes through 0^n , its predecessor and successor are distinct unit vectors e_i, e_j . Replace

$$e_i, 0^n, e_j \quad \text{by} \quad e_i, e_i + e_j, e_j.$$

For $n \geq 3$, the new middle vertex is neither deleted antipode. The complementary replacement handles a geodesic through 1^n . The length is unchanged, so a Hamming geodesic avoiding the deleted vertices exists.

In any coordinate, the two projected sections are $Q_{n-1} - \{0^{n-1}\}$ and $Q_{n-1} - \{1^{n-1}\}$. Neither contains the other. Thus Q_n^{--} is periphery-free and lies outside $\mathcal{R}_{\text{per}}$.

An elementary restriction is Q_{n-1}^- , and an elementary contraction is Q_{n-1} . Hypercubes belong to $\mathcal{R}_{\text{per}}$, as do the graphs Q_m^- : Polat proves the stronger faithful-treelike property [Pol12, Proposition 5.8]. Directly, the same membership follows by expanding Q_{m-1} along Q_{m-1}^- , starting with $Q_1^- = K_1$. Hence every immediate proper pc-minor of Q_n^{--} belongs to $\mathcal{R}_{\text{per}}$; lemma 5.1 puts every further proper minor there. This proves minimality. Any excluded list must therefore contain every Q_n^{--} . $\square$

Knauer and Marc proved that the ample partial cubes are exactly those with no pc-minor Q_n^{--} , $n \geq 3$ [KM20, Corollary 7.4]. It remains to describe the ample members of the basis.

Proposition 6.5 (Structure of a minimal obstruction). *Every member $O \in \mathcal{O}_{\text{per}}$ is either Q_n^{--} for some $n \geq 3$, or is ample. Moreover, every member of $\mathcal{O}_{\text{per}}$ is prime with respect to the Cartesian product.*

Proof. If O is not ample, the characterization of Knauer and Marc gives a pc-minor Q_n^{--} . By proposition 6.4 this minor is outside $\mathcal{R}_{\text{per}}$, so minimality forces $O \cong Q_n^{--}$. If $O = G \square H$ with both factors nontrivial, each factor is a proper convex subgraph and hence a proper pc-minor of O . Thus $G, H \in \mathcal{R}_{\text{per}}$, and proposition 8.1(ii) would give $O \in \mathcal{R}_{\text{per}}$, a contradiction. $\square$

7 Relations with established classes

The position of $\mathcal{R}_{\text{per}}$ is easiest to read one comparison at a time. In the table, the separating example lies in the larger class but not in the smaller one.

Comparison class	Position relative to $\mathcal{R}_{\text{per}}$ and separating example
median graphs	Properly contained in $\mathcal{R}_{\text{per}}$; witness Q_3^- , a daisy cube with no median for three vertices.
daisy cubes	Properly contained in $\mathcal{R}_{\text{per}}$; witness P_4 , whose middle Θ -class is not peripheral.
tree-like partial cubes	Properly contain $\mathcal{R}_{\text{per}}$; witness the sequential bad-site expansion in example 3.3.
almost-median graphs	Properly contain $\mathcal{R}_{\text{per}}$; witness the pentagonal obstruction $\mathcal{P}_5$, proposition 9.1
proper semi-median graphs	Properly contain $\mathcal{R}_{\text{per}}$; witness the same $\mathcal{P}_5$.
corner-peelable ample partial cubes	Properly contain $\mathcal{R}_{\text{per}}$; witness the same $\mathcal{P}_5$.

Proposition 7.1 (Class containments). *For finite partial cubes, the following statements hold.*

- (i) *Every median graph and every daisy cube belongs to $\mathcal{R}_{\text{per}}$, and both containments are proper.*
- (ii) *Every member of $\mathcal{R}_{\text{per}}$ is tree-like, ample, corner-peelable, almost-median, and proper semi-median.*

Proof. Mulder’s peripheral form of the median expansion theorem gives a recursive peripheral decomposition in which the contraction base and duplicated convex site are median; see [Gol14, p. 140]. Hence every median graph belongs to $\mathcal{R}_{\text{per}}$.

For a daisy cube, orient the coordinates so that its vertex family $D \subseteq 2^X$ is a down-set. Sections and projections of a down-set are again down-sets. Moreover, for every nonconstant coordinate x , every member of the $x = 1$ section has its $x = 0$ neighbor, so the 1-semicube is peripheral. Thus every nontrivial pc-minor of a daisy cube has a peripheral class, and [theorem 5.2](#) gives the inclusion. It is proper because the path

$$\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}$$

belongs to $\mathcal{R}_{\text{per}}$, while its middle coordinate is not peripheral; every coordinate of a daisy cube is peripheral. The median inclusion is proper because $Q_3^- = Q_3 - \{111\}$ is the peripheral expansion of Q_2 along a three-vertex path and hence lies in $\mathcal{R}_{\text{per}}$, but the three neighbors of the missing vertex have no median in Q_3^- .

For (ii), forgetting the decompositions on duplicated sites leaves a sequential peripheral construction, so every member is tree-like. We prove ampleness directly by induction. Suppose

$$C = (B \times \{0\}) \cup (S \times \{1\})$$

is a peripheral expansion, and let e be the new coordinate. A set of old coordinates is shattered, respectively strongly shattered, by C exactly when it has that property in B . A coordinate set containing e is shattered, respectively strongly shattered, exactly when its old coordinates have that property in S . Thus

$$\begin{aligned} \text{str}(C) &= \text{str}(B) \dot{\cup} \{T \cup \{e\} : T \in \text{str}(S)\}, \\ \text{sstr}(C) &= \text{sstr}(B) \dot{\cup} \{T \cup \{e\} : T \in \text{sstr}(S)\}. \end{aligned}$$

Equality of shattering and strong-shattering complexes therefore passes from B, S to C .

For corner peeling, take prefix-isometric orderings $b_1, \dots, b_p$ of B and $s_1, \dots, s_q$ of S . The ordering

$$(b_1, 0), \dots, (b_p, 0), (s_1, 1), \dots, (s_q, 1)$$

is prefix-isometric in C . Prefixes in the first block are isometric by induction. A later prefix has the form $B \times \{0\} \cup T \times \{1\}$, where T is an isometric prefix of S . Geodesics within a layer remain there, and a B -geodesic from b to t , followed by the new-coordinate edge, has Hamming length between $(b, 0)$ and $(t, 1)$. The ordering is therefore a corner peeling by [CCMW22, Proposition 4.2].

The cycle $C_6 = Q_3^{--}$ is periphery-free, so [theorem 5.2](#) implies that no member of $\mathcal{R}_{\text{per}}$ has a C_6 pc-minor. Klavžar and Shpectorov characterized the almost-median graphs as the partial cubes with no convex cycle of length at least six [KS12, Theorem 4.1]. Marc’s equivalent pc-minor formulation says that these are exactly the partial cubes with no C_6 pc-minor [Mar18, Theorem 4.4.4]. Hence every member of $\mathcal{R}_{\text{per}}$ is almost-median. Finally, induct through a recursive peripheral decomposition. A peripheral expansion is connected, both cover members are proper semi-median by induction, and the resulting graph is almost-median and therefore semi-median. This is precisely the recursive definition of a proper semi-median graph. $\square$

The two lower comparison classes are incomparable with one another. Their intersection inside the partial cubes is already known:

$$\{\text{median graphs}\} \cap \{\text{daisy cubes}\} = \{\text{simplex graphs}\} [\text{XX25}, \text{Theorem 1.2}].$$

Thus $\mathcal{R}_{\text{per}}$ contains both branches and their intersection, rather than being an intermediate class between median and daisy cubes. The surrounding chain

$$\{\text{corner-peelable ample partial cubes}\} \subsetneq \{\text{ample partial cubes}\} \subsetneq \{\text{almost-median graphs}\}$$

is also strict. An ample partial cube has no C_6 pc-minor by the ample forbidden-minor theorem, so Marc’s criterion gives the second inclusion. Hall’s maximum class is ample but has no corner [CCMW22, Theorem 4.5]. On the other side, Q_4^- is nonample by proposition 6.4, while every proper pc-minor of it belongs to $\mathcal{R}_{\text{per}}$; hence it has no C_6 pc-minor and is almost-median by the characterization used in the proof of proposition 7.1.

Figure 1 summarizes these inclusions together with the larger comparison classes from proposition 7.1.

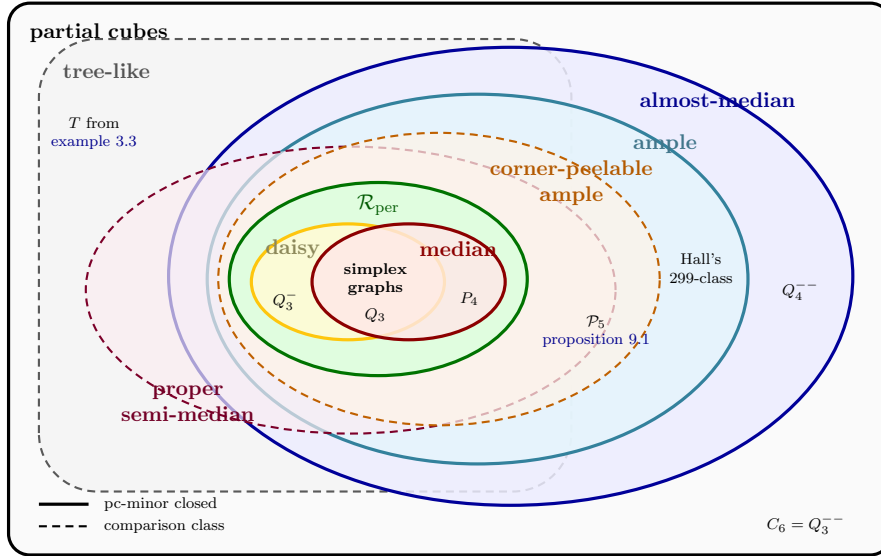


Figure 1: Venn-style containment diagram for the classes used in the paper. Solid outlines mark pc-minor-closed classes; dashed outlines mark comparison classes for which no pc-minor-closure assertion is made here. Complete enclosure records a proved containment. Intersections of nonnested outer outlines are schematic and assert no further relationship. The labels T and $\mathcal{P}_5$ witness the strict containments of $\mathcal{R}_{\text{per}}$ in the indicated outer comparison classes. Hall’s maximum class and Q_4^- witness the two strict containments in the surrounding corner-peelable–ample–almost-median chain.

8 Closure properties and inherited consequences

The hereditary characterizations already give pc-minor, convex-subgraph, and gated-subgraph closure. The following proposition records the remaining elementary operations and the exact limitations.

Proposition 8.1 (Elementary closure properties). *For finite partial cubes, the following statements hold.*

- (i) *A specified peripheral expansion of B along an isometric site S belongs to $\mathcal{R}_{\text{per}}$ if and only if both B and S do.*
- (ii) *$G \square H \in \mathcal{R}_{\text{per}}$ if and only if $G, H \in \mathcal{R}_{\text{per}}$.*
- (iii) *$\mathcal{R}_{\text{per}}$ is closed under convex and gated subgraphs, but not under arbitrary isometric or arbitrary induced subgraphs.*

Proof. For (i), necessity follows because B is a contraction and S is a semicube restriction of the output; sufficiency is the defining closure rule.

For (ii), use induction on $\text{idim}(G) + \text{idim}(H)$. If G is the peripheral expansion of B along S , then $G \square H$ is the peripheral expansion of $B \square H$ along $S \square H$, and both smaller products recursively qualify. The base case is a singleton factor, for which the product is the other factor; if only H is nontrivial, swap the factors. Conversely, each factor is a convex layer of the product and hence belongs to $\mathcal{R}_{\text{per}}$ by [lemma 5.1](#).

Convex and gated closure in (iii) follows from [lemma 5.1](#). For the negative assertions, the hypercube Q_3 is median and hence belongs to $\mathcal{R}_{\text{per}}$, but the middle-level $C_6 = Q_3^{--}$ is an induced isometric subgraph and is periphery-free. $\square$

Remark 8.2 (What heredity is and is not obtained). The valid chain for subgraphs of a fixed graph is

$$\text{gated} \implies \text{convex} \implies \text{isometric} \implies \text{induced}.$$

[Lemma 5.1](#) gives convex-subgraph closure and therefore also gated-subgraph closure. It does not give closure under the two weaker notions, as [proposition 8.1\(iii\)](#) shows. This differs from the standard tree-like class, for which gated-subgraph closure is valid but convex-subgraph closure fails [[Gol14](#)].

Several further properties follow from known results for the tree-like or ample classes. The *cubical complex* of a coordinate family is formed by filling every contained cube, including all its faces. An elementary cubical collapse removes a maximal cube together with a facet contained in no other maximal cube. A complex is *collapsible* if a sequence of such collapses leaves one vertex.

The intersection graph of the maximal cubes has those cubes as vertices, with adjacency when they meet. For this graph, *dismantlable* means reducible to one vertex by repeatedly deleting a vertex u whose closed neighborhood is contained in that of another vertex v . This is the graph-theoretic meaning of dismantlability, distinct from a corner peeling of the original partial cube.

Proposition 8.3 (Inherited structural properties). *Let $G \in \mathcal{R}_{\text{per}}$. Then:*

- (i) *if G is regular, then G is a hypercube;*
- (ii) *the intersection graph of the maximal cubes of G is dismantlable;*
- (iii) *if $\alpha_i(G)$ is the number of induced i -cubes, then $\sum_{i \geq 0} (-1)^i \alpha_i(G) = 1$; and*
- (iv) *the cubical complex of G is collapsible, and hence contractible.*

Proof. The first three conclusions hold for every tree-like partial cube [[BIK03b](#)]; the corrected proof of (ii) is [[Gol14](#), Corollary 15]. They therefore follow from [proposition 7.1\(ii\)](#). The specialization of (i) to convex-treelike graphs is also [[Pol12](#), Proposition 5.9]. For (iv), ampleness from [proposition 7.1\(ii\)](#) suffices: the cubical complex of every ample class is collapsible [[CCMW22](#), Proposition 4.12]. The corner peeling also gives such a sequence directly, as described in Section 4.4 of that source. $\square$

Remark 8.4 (Weak retracts). Gologranc identified a gap in the original proof of weak-retract closure for tree-like partial cubes and left that question open in [Gol14]. The closure results here concern convex subgraphs and coordinate quotients.

9 Explicit ample obstructions

The nonample obstruction family was identified in [proposition 6.4](#). We now give explicit ample members of the second family in [theorem 11.3](#), beginning with the smallest one beyond the punctured cubes.

The following table provides a quick gallery before the general two-layer formalism. The first three rows are globally ordered: there is no obstruction of order below six, Q_4^{--} is the next nonample member, and $\mathcal{P}_5$ is the smallest obstruction outside the punctured-cube family. The later rows are small named examples, not a claim that all intervening orders have been classified.

Obstruction	Order	idim	Description
$Q_3^{--} \cong C_6$	6	3	first obstruction; nonample and periphery-free
Q_4^{--}	14	4	second antipodally punctured cube; nonample
$\mathcal{P}_5$	16	5	smallest additional obstruction; maximum of VC dimension two, proposition 9.1 and theorem 9.14
$\mathcal{A}_n, n \geq 5$	$3n + 1$	n	cyclic arcs of length at most three; infinite ample family with $\mathcal{A}_5 = \mathcal{P}_5$, theorem 9.4
$\mathcal{F}_n, n \geq 9$ odd	$10n + 1$	n	VC dimension three; peeled in steps of two, theorem 9.7
$\mathcal{S}_{18}$	18	6	ample square-amalgam obstruction with no cut vertex, proposition 9.11
$\mathcal{W}_8$	21	8	ample one-vertex amalgam of two root-critical factors, proposition 10.7
$\mathcal{H}_{22}$	22	6	ample obstruction extracted from Hall's maximum class, proposition 9.12

9.1 The pentagonal obstruction

Let C_5 be the cycle on $[5] = \{0, 1, 2, 3, 4\}$, with indices modulo five, and define

$$\mathcal{P}_5 = \{\emptyset\} \cup \binom{[5]}{1} \cup E(C_5) \cup \{[5] \setminus e : e \in E(C_5)\} \subseteq 2^{[5]}. \quad (5)$$

Thus $\mathcal{P}_5$ has sixteen vertices.

Proposition 9.1 (Pentagonal obstruction). *The graph $\mathcal{P}_5$ is a maximum VC-dimension-two ample partial cube, has a corner peeling, and is proper semi-median. Moreover,*

$$\mathcal{P}_5 \in \mathcal{O}_{\text{per}} \setminus \{Q_n^{--} : n \geq 3\}.$$

Proof. Every coordinate pair is shattered. For an edge of C_5 , the two-one pattern is supplied by that edge; for a nonedge, choose an edge disjoint from the pair and use its complement. The empty set and singletons supply the other three patterns. No triple T is shattered. If $[5] \setminus T$ is not an edge, no member realizes the all-one pattern on T . If it is an edge, the vertices of T induce a path; the pattern selecting the two endpoints and omitting the middle vertex is absent. Hence

$$\dim_{\text{VC}}(\mathcal{P}_5) = 2, \quad |\mathcal{P}_5| = 1 + 5 + \binom{5}{2}.$$

The family is maximum, hence ample and a partial cube. Every ample class of VC dimension at most two has a corner peeling [CCMW22, Proposition 4.11].

For every coordinate, the projected sections have orders ten and six, their intersection has order five, and neither contains the other. By cyclic symmetry, take coordinate 0: the singleton $\{2\}$ occurs only in the zero-section, whereas the nonedge $\{1, 4\}$ occurs only in the one-section. Thus every coordinate is nonperipheral and $\mathcal{P}_5 \notin \mathcal{R}_{\text{per}}$.

It remains to verify pc-minimality. Cyclic symmetry leaves three types of elementary minor: the two restrictions, of orders six and ten, and the contraction, of order eleven. Up to cube symmetry their vertex families are

$$\begin{aligned} H_6 &= \{0000, 0001, 0010, 0011, 0111, 1000\}, \\ H_{10} &= \{0000, 0001, 0010, 0011, 0101, 0110, 0111, 1000, 1001, 1010\}, \\ H_{11} &= \{0000, 0001, 0010, 0011, 0100, 0110, 0111, 1000, 1001, 1011, 1100\}. \end{aligned}$$

Appendix A gives an explicit recursive decomposition of each of these three families. Thus all fifteen elementary pc-minors belong to $\mathcal{R}_{\text{per}}$. By lemma 5.1, so does every proper pc-minor, and $\mathcal{P}_5 \in \mathcal{O}_{\text{per}}$.

Finally, in each coordinate the overlap of the two sections is a five-vertex path. Both sections and their contraction belong to $\mathcal{R}_{\text{per}}$ by the decompositions in appendix A and hence are proper semi-median by proposition 7.1. The output $\mathcal{P}_5$ is ample and therefore almost-median, hence semi-median. Its connected expansion across any coordinate satisfies the recursive definition of proper semi-median [BIK⁺02, Section 3, class (10)]. $\square$

Corollary 9.2 (Sharpness of the position statements). *The containments*

$$\mathcal{R}_{\text{per}} \subsetneq \{\text{tree-like}\} \cap \{\text{proper semi-median}\}$$

and

$$\mathcal{R}_{\text{per}} \subsetneq \{\text{tree-like}\} \cap \{\text{almost-median}\} \cap \{\text{ample and corner-peelable}\}$$

are strict.

Proof. Form the peripheral expansion

$$X = (Q_5 \times \{0\}) \cup (\mathcal{P}_5 \times \{1\}).$$

It is tree-like. The overlap $\mathcal{P}_5$ is isometric, and the two cover members $Q_5, \mathcal{P}_5$ are almost-median, so the one-step preservation theorem makes X almost-median [BIK03a, Theorem 4]. The same cover witnesses that X is proper semi-median: both members are recursively proper semi-median, the overlap is connected, and the output is semi-median.

The shattering calculation in the proof of proposition 7.1 shows that a peripheral expansion of an ample base along an ample isometric site is ample. The concatenated prefix-isometric ordering there gives a corner peeling. However, $\mathcal{P}_5 \times \{1\}$ is a convex semicube of X . If $X \in \mathcal{R}_{\text{per}}$, convex-subgraph closure would put $\mathcal{P}_5$ in $\mathcal{R}_{\text{per}}$, a contradiction. $\square$

9.2 An infinite family of cyclic-arc obstructions

The pentagonal class has a second description. Index the coordinates by $\mathbb{Z}_5$. An edge $\{i, i+1\}$ of C_5 is a cyclic arc of length two, and its complement is the cyclic arc $\{i+2, i+3, i+4\}$ of length three. Thus $\mathcal{P}_5$ consists of the empty set and all cyclic arcs of length at most three. Keeping the arc length fixed while the cycle grows gives an infinite family of ample obstructions.

For $n \geq 4$ and $a \in \mathbb{Z}_n$, write $[a, a + \ell) = \{a, a + 1, \dots, a + \ell - 1\} \subseteq \mathbb{Z}_n$ for the cyclic arc of length ℓ , and put

$$\mathcal{A}_n = \{\emptyset\} \cup \{[a, a + \ell) : a \in \mathbb{Z}_n, 1 \leq \ell \leq 3\} \subseteq 2^{\mathbb{Z}_n}. \quad (6)$$

The $3n$ arcs are distinct, so $|\mathcal{A}_n| = 3n + 1$, and $\mathcal{A}_5 = \mathcal{P}_5$. For the proof we also use the path-interval families

$$\mathcal{J}_m = \{\emptyset\} \cup \{[a, \dots, a + \ell - 1] : 1 \leq a, a + \ell - 1 \leq m, 1 \leq \ell \leq 3\} \subseteq 2^{[m]}, \quad m \geq 0. \quad (7)$$

Lemma 9.3 (Path intervals). *Every $\mathcal{J}_m$ belongs to $\mathcal{R}_{\text{per}}$.*

Proof. Induct on m ; $\mathcal{J}_0 = \{\emptyset\}$ is a singleton. For $m \geq 1$, the members of $\mathcal{J}_m$ containing m are $\{m\}, \{m - 1, m\}, \{m - 2, m - 1, m\}$, as far as these are defined. Deleting m gives $\emptyset, \{m - 1\}, \{m - 2, m - 1\}$, which are members of $\mathcal{J}_m$ avoiding m . Hence the 1-semicube of coordinate m is peripheral. Its contraction base is the zero section $\mathcal{J}_{m-1}$, and its duplicated site is a path with at most three vertices. The base lies in $\mathcal{R}_{\text{per}}$ by induction, and the site is a tree and hence lies in $\mathcal{R}_{\text{per}}$ by [proposition 7.1](#). $\square$

Theorem 9.4 (Cyclic-arc obstructions). *For every $n \geq 5$, the family $\mathcal{A}_n$ is an ample partial cube of VC dimension two and isometric dimension n with $3n + 1$ vertices, and*

$$\mathcal{A}_n \in \mathcal{O}_{\text{per}} \setminus \{Q_m^{--} : m \geq 3\}.$$

Proof. Shattering. The empty set and the singletons supply the patterns 00, 10, 01 on every pair $\{i, j\}$. The pattern 11 is realized exactly when some arc of length at most three contains both points, that is, when their cyclic distance is one or two. Hence $\text{str}(\mathcal{A}_n)$ contains exactly $2n$ pairs. No triple is shattered. A triple with the pattern 111 lies in an arc of length three, so it is $\{i, i + 1, i + 2\}$. An arc containing i and $i + 2$ but not $i + 1$ runs the other way around the cycle and has length at least $n - 1 \geq 4$. Therefore

$$|\text{str}(\mathcal{A}_n)| = 1 + n + 2n = |\mathcal{A}_n|,$$

and $\mathcal{A}_n$ has VC dimension two. Every shattered set is strongly shattered: $\{i\}$ by the edge from $\emptyset$ to $\{i\}$; $\{i, i + 1\}$ by the square $\emptyset, \{i\}, \{i + 1\}, \{i, i + 1\}$; and $\{i, i + 2\}$ by the square $\{i + 1\}, \{i, i + 1\}, \{i + 1, i + 2\}, \{i, i + 1, i + 2\}$. Thus $\mathcal{A}_n$ is ample [\[BCDK06\]](#). Ample families induce isometric subgraphs of the cube [\[BCDK06\]](#), so $\mathcal{A}_n$ is a partial cube. Every coordinate is nonconstant, so its isometric dimension is n .

No peripheral coordinate. Rotation acts transitively on the coordinates, so consider coordinate 0. Its projected sections are

$$S^1 = \{x \setminus \{0\} : 0 \in x \in \mathcal{A}_n\}, \quad S^0 = \{x \in \mathcal{A}_n : 0 \notin x\}.$$

Since $n \geq 5$, the singleton $\{3\}$ lies in S^0 but not in S^1 , whereas $\{n - 1, 1\} = [n - 1, n + 2) \setminus \{0\}$ lies in S^1 but is not an arc avoiding 0. Neither section contains the other, so $\mathcal{A}_n$ has no peripheral coordinate and $\mathcal{A}_n \notin \mathcal{R}_{\text{per}}$.

Minimality. Again consider coordinate 0. The zero section S^0 consists of the empty set and the arcs avoiding 0. These are the intervals of length at most three in the path $1, 2, \dots, n - 1$, so $S^0 \cong \mathcal{J}_{n-1} \in \mathcal{R}_{\text{per}}$ by [lemma 9.3](#).

The one section is

$$S^1 = \{\emptyset, \{1\}, \{n - 1\}, \{1, 2\}, \{n - 2, n - 1\}, \{n - 1, 1\}\}.$$

It is the square on $\emptyset, \{1\}, \{n-1\}, \{n-1, 1\}$ with pendant vertices $\{1, 2\}$ and $\{n-2, n-1\}$; the four active coordinates $n-2, n-1, 1, 2$ are distinct because $n \geq 5$. The 1-semicube of coordinate 2 is the single vertex $\{1, 2\}$, whose neighbor $\{1\}$ is present. It is therefore peripheral, with a singleton site. Removing it, and then removing $\{n-2, n-1\}$ in the same way, leaves a square. Thus $S^1 \in \mathcal{R}_{\text{per}}$.

The contraction $K = \mathcal{A}_n/0$ consists of the empty set, the arcs avoiding 0, and the sets

$$\emptyset, \{1\}, \{n-1\}, \{1, 2\}, \{n-2, n-1\}, \{n-1, 1\}$$

obtained by deleting 0 from arcs through it. All of these except $\{n-1, 1\}$ already avoid 0. In K , consider coordinate 1. The members containing 1 are $\{1\}, \{1, 2\}, \{1, 2, 3\}, \{n-1, 1\}$. Deleting 1 gives $\emptyset, \{2\}, \{2, 3\}, \{n-1\}$, and these are members of K avoiding 1. Hence the 1-semicube of coordinate 1 is peripheral. Its duplicated site is the path $\{n-1\}, \emptyset, \{2\}, \{2, 3\}$, which lies in $\mathcal{R}_{\text{per}}$. Its contraction base consists of the members of K avoiding 1. These are the intervals of length at most three in the path $2, 3, \dots, n-1$, together with the empty set, so the base is isomorphic to $\mathcal{J}_{n-2} \in \mathcal{R}_{\text{per}}$. Thus $K \in \mathcal{R}_{\text{per}}$.

By rotation, every elementary restriction and contraction of $\mathcal{A}_n$ belongs to $\mathcal{R}_{\text{per}}$. [Lemma 5.1](#) puts every proper pc-minor in $\mathcal{R}_{\text{per}}$, so $\mathcal{A}_n \in \mathcal{O}_{\text{per}}$. Since $\mathcal{A}_n$ is ample, it is not a punctured cube Q_m^{--} . $\square$

Remark 9.5 (Why the arc length is three). With arcs of length at most two, the family is a daisy cube: it is a down-set in $2^{\mathbb{Z}_n}$. It therefore lies in $\mathcal{R}_{\text{per}}$ by [proposition 7.1](#). With arcs of length at most four, the family is not a partial cube for $n \leq 6$. For $n \geq 7$ its contraction is the family of arcs of length at most four on $\mathbb{Z}_{n-1}$, with the three arcs of length four through one edge missing. An exact check for $7 \leq n \leq 9$ finds this contraction outside $\mathcal{R}_{\text{per}}$. The maximum classes of VC dimension two formed by the empty set and all cyclic arcs of length at most $(n+1)/2$, for odd n , agree with $\mathcal{P}_5$ at $n = 5$. They are not obstructions for $n = 7, 9$. Thus the fixed length three is exactly what makes the pentagonal obstruction extend to all $n \geq 5$. The same check confirms [theorem 9.4](#) directly for $5 \leq n \leq 13$.

9.3 A family of VC dimension three peeled in steps of two

The second family has VC dimension three and exists exactly on cycles of odd length in the range we tested. Its proof shows how the parity enters: the recursive peeling of a minor advances around $\mathbb{Z}_n$ in steps of two, and it reaches every coordinate only when 2 generates $\mathbb{Z}_n$.

For $n \geq 9$, call a nonempty set $S \subseteq \mathbb{Z}_n$

- *short* if $|S| \leq 3$ and $S \subseteq [a, a+4)$ for some a ;
- *long* if $\{a, a+2, a+4\} \subseteq S \subseteq [a, a+5)$ and $|S| \leq 4$ for some a , called its *left end*; then $a+2$ is its *midpoint*.

Put

$$\mathcal{F}_n = \{\emptyset\} \cup \{S : S \text{ short or long}\} \subseteq 2^{\mathbb{Z}_n}. \quad (8)$$

Up to translation, the nonempty members are

$$\{0\}, \{0, 1\}, \{0, 2\}, \{0, 3\}, \{0, 1, 2\}, \{0, 1, 3\}, \{0, 2, 3\}, \{0, 2, 4\}, \{0, 1, 2, 4\}, \{0, 2, 3, 4\}.$$

Since $n \geq 9$, every nonempty member lies in an arc of length at most five, and this arc is unique; hence these ten types give $10n$ distinct members and $|\mathcal{F}_n| = 10n + 1$. Equivalently, $\mathcal{F}_n$ is the union of $\{\emptyset\}$ with the dihedral orbits of $\{0\}, \{0, 1\}, \{0, 2\}, \{0, 3\}, \{0, 1, 2\}, \{0, 1, 3\}, \{0, 2, 4\}, \{0, 1, 2, 4\}$.

Lemma 9.6 (Deleting one point). *Let $n \geq 9$, let $S \in \mathcal{F}_n$, and let $x \in S$. Then $S \setminus \{x\} \in \mathcal{F}_n$, unless S is long with midpoint x . In the exceptional case $S \setminus \{x\} \notin \mathcal{F}_n$.*

Proof. A nonempty subset of a short set is short. Let S be long with left end a . Deleting $a + 1$ or $a + 3$ leaves the long set $\{a, a + 2, a + 4\}$. Deleting a leaves at most three points of $[a + 1, a + 5)$, a short set, and deleting $a + 4$ is symmetric. If $x = a + 2$, then $S \setminus \{x\}$ contains a and $a + 4$ but not $a + 2$. Its unique arc of length at most five is $[a, a + 5)$, so it is neither short nor long. $\square$

Theorem 9.7 (Step-two obstructions). *For every odd $n \geq 9$, the family $\mathcal{F}_n$ is an ample partial cube of VC dimension three and isometric dimension n with $10n + 1$ vertices, and*

$$\mathcal{F}_n \in \mathcal{O}_{\text{per}} \setminus \{Q_m^{--} : m \geq 3\}.$$

In every coordinate one of the two one-sided section defects has exactly three vertices, and these three vertices induce a path.

The proof has three steps. Ampleness is a count of shattered sets. For minimality it suffices, by rotation, to treat coordinate 0. Its one-section is a fixed 27-vertex family L . Its zero-section and its contraction are peeled one coordinate at a time along 2, 4, 6, $\dots$; lemma 9.6 makes each of these coordinates peripheral, and every duplicated site is a pc-minor of L .

Proof. Shattering. If $\mathcal{F}_n$ shatters T , then T is contained in a member. Up to translation and reflection, the subsets of members with at least two points are the pairs $\{0, d\}$, $1 \leq d \leq 4$, the triples

$$\{0, 1, 2\}, \{0, 1, 3\}, \{0, 1, 4\}, \{0, 2, 4\},$$

and the two long quadruples. The triple $\{i, i + 2, i + 4\}$ is not shattered: a member containing i and $i + 4$ is long with left end i , and therefore contains $i + 2$. Consequently neither long quadruple is shattered. Every other listed set is strongly shattered. The pairs $\{i, i + d\}$ with $d \leq 3$ and the triples $\{i, i + 1, i + 2\}$, $\{i, i + 1, i + 3\}$, $\{i, i + 2, i + 3\}$ have all their subsets short, so they span a cube at $\emptyset$. The sets $\{i, i + 4\}$, $\{i, i + 1, i + 4\}$, and $\{i, i + 3, i + 4\}$ span cubes at $\{i + 2\}$: each union of $\{i + 2\}$ with one of their subsets is short or long. Counting the empty set, singletons, $4n$ pairs, and $5n$ triples gives

$$|\text{sstr}(\mathcal{F}_n)| = |\text{str}(\mathcal{F}_n)| = 1 + n + 4n + 5n = |\mathcal{F}_n|.$$

Hence $\mathcal{F}_n$ is ample of VC dimension three [BCDK06], and in particular it is a partial cube. All n coordinates are nonconstant.

No peripheral coordinate. Rotations are automorphisms, so consider coordinate 0, with sections

$$S^1 = \{U : U \cup \{0\} \in \mathcal{F}_n, 0 \notin U\}, \quad S^0 = \{U \in \mathcal{F}_n : 0 \notin U\}.$$

By lemma 9.6,

$$S^1 \setminus S^0 = \{\{-2, 2\}, \{-2, -1, 2\}, \{-2, 1, 2\}\},$$

the three long sets with midpoint 0 after deleting 0. They induce the path

$$\{-2, -1, 2\}, \{-2, 2\}, \{-2, 1, 2\}.$$

In the other direction, $\{5\} \in S^0$, whereas $\{0, 5\}$ is not a member: a two-point member is short, so its points have cyclic distance at most three. Thus coordinate 0 is not peripheral, and $\mathcal{F}_n \notin \mathcal{R}_{\text{per}}$.

The one-section. The family $L = S^1$ depends only on the arc $[-4, 5)$, whose nine points are distinct. Its 27 members are the empty set, the six singletons $\{\pm 1\}, \{\pm 2\}, \{\pm 3\}$, the nine short pairs

$$\{-3, -2\}, \{-3, -1\}, \{-2, -1\}, \{-2, 1\}, \{-1, 1\}, \{-1, 2\}, \{1, 2\}, \{1, 3\}, \{2, 3\},$$

and the eleven sets coming from long members through 0:

$$\begin{aligned} &\{2, 4\}, \{1, 2, 4\}, \{2, 3, 4\}, \{-4, -2\}, \{-4, -3, -2\}, \{-4, -2, -1\}, \\ &\{-2, 2\}, \{-2, -1, 2\}, \{-2, 1, 2\}, \{-1, 1, 3\}, \{-3, -1, 1\}. \end{aligned}$$

The members containing 4 are $\{2, 4\}, \{1, 2, 4\}, \{2, 3, 4\}$; deleting 4 gives members, so the 1-semicube of 4 is peripheral with a three-vertex path as site. The coordinate -4 is symmetric. In the remaining family, the members containing 3 are

$$\{3\}, \{1, 3\}, \{2, 3\}, \{-1, 1, 3\};$$

deleting 3 gives the tree $\emptyset, \{1\}, \{2\}, \{-1, 1\}$, all of whose vertices remain. The coordinate -3 is symmetric. In the 13-member family left on $\{-2, -1, 1, 2\}$, the members containing 2 are

$$\{2\}, \{-1, 2\}, \{1, 2\}, \{-2, 2\}, \{-2, -1, 2\}, \{-2, 1, 2\}.$$

Deleting 2 gives

$$\{\emptyset, \{-2\}\} \times \{\emptyset, \{-1\}, \{1\}\},$$

a median graph, and all six vertices remain. What is left is $2^{\{-2, -1, 1\}} \setminus \{\{-2, -1, 1\}\}$, a cube with one vertex deleted, which belongs to $\mathcal{R}_{\text{per}}$ by the proof of [proposition 6.4](#). Trees and median graphs belong to $\mathcal{R}_{\text{per}}$ by [proposition 7.1](#). Each displayed peripheral step reconstructs the previous family, so $L \in \mathcal{R}_{\text{per}}$. By rotation, the one-section $L_p = \{U : U \cup \{p\} \in \mathcal{F}_n, p \notin U\}$ at every coordinate p is isomorphic to L .

Peeling in steps of two. For $Z \subseteq \mathbb{Z}_n$ and an optional point $c \notin Z$, put

$$H(Z; c) = \{S \setminus \{c\} : S \in \mathcal{F}_n, S \cap Z = \emptyset\},$$

and write $H(Z)$ when no point is contracted. These are pc-minors of $\mathcal{F}_n$: restrict every coordinate of Z to zero and contract c . They are therefore partial cubes.

Claim. Let $p \notin Z \cup \{c\}$ and $p - 2 \in Z \cup \{c\}$. Then in $H = H(Z; c)$ the 1-semicube of p is peripheral. Its zero-section is $H(Z \cup \{p\}; c)$, and its duplicated site lies in $\mathcal{R}_{\text{per}}$.

Let $T \in H$ contain p , say $T = S \setminus \{c\}$ with $S \in \mathcal{F}_n$ avoiding Z . If $S \setminus \{p\} \in \mathcal{F}_n$, then $T \setminus \{p\} = (S \setminus \{p\}) \setminus \{c\} \in H$. Otherwise, by [lemma 9.6](#), S is long with midpoint p , so $p - 2 \in S$. As S avoids Z , this forces $p - 2 = c$. Then $T \setminus \{p\}$ is $\{p + 2\}, \{p - 1, p + 2\}$, or $\{p + 1, p + 2\}$. Each is a short member of $\mathcal{F}_n$ avoiding $Z \cup \{c\}$, so it lies in H . Thus the one-section of p is contained in its zero-section, which proves peripherality. The zero-section is $H(Z \cup \{p\}; c)$ by definition. The duplicated site is

$$\{S \setminus \{c, p\} : p \in S \in \mathcal{F}_n, S \cap Z = \emptyset\},$$

obtained from L_p by restricting the coordinates in Z to zero and contracting c . It is a pc-minor of $L_p \cong L$, so it lies in $\mathcal{R}_{\text{per}}$ by [lemma 5.1](#). Also $\{p\} \in H$, so the coordinate is nonconstant. This proves the claim.

Because n is odd, the points $0, 2, 4, \dots, 2(n-1)$ are all distinct. For the zero-section $S^0 = H(\{0\})$, put

$$Z_t = \{0, 2, \dots, 2t - 2\}, \quad p_t = 2t \quad (1 \leq t \leq n - 1).$$

Then $p_t \notin Z_t$ and $p_t - 2 \in Z_t$. The claim writes $H(Z_t)$ as a peripheral expansion of $H(Z_{t+1})$ along a site in $\mathcal{R}_{\text{per}}$. Since $Z_n = \mathbb{Z}_n$, the last family $H(Z_n)$ is the singleton $\{\emptyset\}$. Descending induction on t gives $S^0 = H(Z_1) \in \mathcal{R}_{\text{per}}$.

For the contraction $\mathcal{F}_n/0 = H(\emptyset; 0)$, put $c = 0$ and

$$Z'_t = \{2, 4, \dots, 2t\}, \quad p'_t = 2t + 2 \quad (0 \leq t \leq n - 2).$$

Then $p'_t \notin Z'_t \cup \{0\}$, and $p'_t - 2$ is c for $t = 0$ and lies in Z'_t otherwise. The claim again gives a chain of peripheral expansions. It ends at $Z'_{n-1} = \mathbb{Z}_n \setminus \{0\}$, where $H(Z'_{n-1}; 0) = \{\emptyset\}$. Hence $\mathcal{F}_n/0 \in \mathcal{R}_{\text{per}}$.

Thus both restrictions and the contraction at 0, and by rotation at every coordinate, belong to $\mathcal{R}_{\text{per}}$. Lemma 5.1 gives $\mathcal{F}_n \in \mathcal{O}_{\text{per}}$. It is ample, hence not a punctured cube. Finally, the defect $S^1 \setminus S^0$ computed above has three vertices inducing a path, and rotation transports this to every coordinate. The opposite defect $S^0 \setminus S^1$ is larger: it contains $\{5\}$, and $|S^0| - |S^1 \cap S^0| = 10n + 1 - 27 - 24 \geq 40$. $\square$

Remark 9.8 (Parity and verification). The proof uses oddness only to make the step-two walks visit every point. For even n they stop after the even coordinates. An exact check for $n = 10, 12, 14$ finds that S^0 and $\mathcal{F}_n/0$ are then outside $\mathcal{R}_{\text{per}}$, so $\mathcal{F}_n$ is not an obstruction. For odd $9 \leq n \leq 51$, a program replays every step of both chains, including each containment and each site. Independently, the full pc-minor recursion confirms $\mathcal{F}_n \in \mathcal{O}_{\text{per}}$ for $n = 9, 11, 13, 15, 17$. The coordinate-state test of theorem 6.1 confirms it for $n = 9$. Theorem 9.7 also shows that single-vertex deletion from a member of $\mathcal{R}_{\text{per}}$ does not produce every ample obstruction. Here, in every coordinate, the smaller one-sided defect has three vertices.

9.4 No criterion of bounded depth

By theorem 6.1, a periphery-free partial cube is an obstruction exactly when every proper nontrivial pc-minor, at every depth, has a peripheral coordinate. The following family shows that no fixed depth suffices. Fix the coordinate 0 of $\mathcal{P}_5$ and its two arcs

$$a_0 = \{1, 2, 3\}, \quad b_0 = \{0, 1, 2\}.$$

Each lies in a unique maximal square of $\mathcal{P}_5$, namely

$$\{2\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\} \quad \text{and} \quad \{1\}, \{0, 1\}, \{1, 2\}, \{0, 1, 2\},$$

and a_0 does not lie in the square of b_0 . For $k \geq 1$, let Y be a set of k new coordinates and put

$$\mathcal{T}_k = (\mathcal{P}_5 \times 2^Y) \setminus \{(a_0, Y), (b_0, \emptyset)\}. \quad (9)$$

The two deleted vertices sit above the antipodal points Y and $\emptyset$ of the auxiliary cube.

Theorem 9.9 (No criterion of bounded depth; computer-assisted base). *For every $k \geq 1$, the family $\mathcal{T}_k$ is an ample partial cube of isometric dimension $k + 5$ with $2^{k+4} - 2$ vertices. It has no peripheral coordinate and does not belong to $\mathcal{O}_{\text{per}}$, since contracting the k coordinates of Y gives $\mathcal{P}_5$. Every nontrivial pc-minor of $\mathcal{T}_k$ obtained by fewer than k elementary operations has a peripheral coordinate.*

Consequently, for every d , the periphery-free partial cubes all of whose pc-minors of depth less than d have a peripheral coordinate form a strictly larger family than $\mathcal{O}_{\text{per}}$. The quantification over all states in theorem 6.1 cannot be truncated.

Proof. Ampleness. The product of ample families is ample, since shattered and strongly shattered sets of a product are the unions of those of the factors. In an ample family, a vertex lies in a unique maximal cube exactly when deleting it leaves an ample family [CCMW22, Lemma 4.1]. Maximal

cubes of $\mathcal{P}_5 \times 2^Y$ are products of maximal cubes, so (a_0, Y) lies only in $M_a \times 2^Y$, where M_a is the square of a_0 . Its deletion leaves an ample family. In that family $(b_0, \emptyset)$ lies only in $M_b \times 2^Y$, which avoids (a_0, Y) because $a_0 \notin M_b$. Deleting it leaves $\mathcal{T}_k$ ample. Ample families are partial cubes, and every coordinate is nonconstant.

Fibres. For a state τ that retains every coordinate of Y , write $F = \mathcal{P}_5[\tau]$, and let F^a and F^b be the corresponding minors of $\mathcal{P}_5 \setminus \{a_0\}$ and $\mathcal{P}_5 \setminus \{b_0\}$. Then $\mathcal{T}_k[\tau]$ has fibre F^a over Y , fibre F^b over $\emptyset$, and fibre F over every other subset of Y . A coordinate of $\mathcal{P}_5$ is peripheral with a given orientation exactly when the corresponding section containment holds in each fibre. An auxiliary coordinate is peripheral exactly when the fibres on one side are contained in the matching fibres on the other. The pairs to compare are (F^b, F) and (F, F^a) when $k \geq 2$, with further pairs (F, F) when $k \geq 3$. Hence, for $k \geq 2$, whether $\mathcal{T}_k[\tau]$ has a peripheral coordinate does not depend on k . An exact computation of all states τ of the five coordinates of $\mathcal{P}_5$, for $k = 1$ and $k = 2$, shows that $\mathcal{T}_k$ itself is periphery-free and that every other state with at least two vertices has a peripheral coordinate. There are 863 such states for $k = 1$ and 917 for $k = 2$.

Words touching Y . Let a word of fewer than k elementary operations apply at least one operation to a coordinate $s \in Y$. Then some $t \in Y$ is untouched. Since operations in distinct coordinates commute, the minor is $\mathcal{T}_k[\sigma]$ for a state σ . If s is contracted, each deleted vertex is replaced by its s -neighbour, which is not the other deleted vertex because t is retained. Thus $\mathcal{T}_k[\sigma] = (\mathcal{P}_5 \times 2^Y)[\sigma]$ is a product with a full factor in t , so t is peripheral. If s is restricted, one deleted vertex lies outside the chosen semicube. The minor is then a product in which one fibre may lack a single point, and the t -section containing that fibre is contained in the other. Hence t is peripheral, unless the minor has at most two vertices, in which case it is a vertex or an edge.

Depth exactly k . Words without an operation in Y are covered by the fibre computation. Contracting all of Y replaces each deleted vertex by a surviving copy and gives $\mathcal{P}_5$, which is periphery-free and a proper pc-minor. Therefore $\mathcal{T}_k \notin \mathcal{R}_{\text{per}}$. It is not minimal, since its proper minor $\mathcal{P}_5$ lies outside $\mathcal{R}_{\text{per}}$. $\square$

Remark 9.10 (Consequences for a generative description). By [proposition 6.3](#), every obstruction is a member of $\mathcal{R}_{\text{per}}$ with one defect deleted. [Theorem 9.9](#) limits what a converse can look like. Periphery tests at bounded depth cannot certify minimality, and the family $\mathcal{P}_5 \times K_2$ shows that adding $\mathcal{R}_{\text{per}}$ -membership of the minors along one coordinate is still not enough. The criteria in [theorem 6.1](#) and [section 11](#) account for this global minimality condition; a structural generator must also account for it, although it need not use a recursive membership predicate. Unlike daisy membership, a fixed-depth periphery test cannot supply the condition.

The earlier finite data suggested a defect bound of three vertices, but this bound is false. Replacing every coordinate in the signed clauses of $\mathcal{P}_5$ by two coordinates with the same forbidden sign gives an ample forbidden minor of order 918, isometric dimension 10, and VC dimension 7, with oriented defects of sizes 9 and 45 at every coordinate. Uniform coordinate-block replacement makes the minimum defect unbounded even at fixed dimension difference three; the [coordinate-block research note](#) contains the construction, full proofs, and direct verification. For one-vertex-decomposable obstructions one vertex suffices in all 47 root-critical factors computed through dimension five; that separate assertion remains open in general.

9.5 The square-amalgam obstruction

In Q_6 , with the leftmost bit denoting coordinate 5, define

$$\begin{aligned} \mathcal{S}_{18} = \{ & 000000, 000100, 001100, 010000, 010100, 100100, 101100, 110000, \\ & 110001, 110011, 110100, 111000, 111001, 111010, 111011, 111100, 111110, 111111 \}. \end{aligned} \quad (10)$$

Proposition 9.11 (Square-amalgam obstruction; computer-assisted). *The family $\mathcal{S}_{18}$ is an ample VC-dimension-two partial cube and*

$$\mathcal{S}_{18} \in \mathcal{O}_{\text{per}}.$$

It has no cut vertex and is the gated amalgam of two copies of $\mathcal{L}_4$ from (12) along a maximal square. Up to cube isomorphism, it is the unique ample periphery-free partial cube of isometric dimension six and order at most eighteen.

Proof. The exact verifier applied to (10) checks the Hamming-distance criterion for a partial cube, Sandwich equality

$$|\mathcal{S}_{18}| = 1 + 6 + 11,$$

the absence of a peripheral orientation in every coordinate, and recursive peripheral expansion trees for all sixteen distinct normalized elementary pc-minors. Pc-minor closure then proves $\mathcal{S}_{18} \in \mathcal{O}_{\text{per}}$. The same computation records a complete corner peeling, although only ampleness is needed here.

For the amalgam description, put

$$A = \{0, 1\}, \quad I = \{2, 3\}, \quad B = \{4, 5\},$$

and

$$L = \{x \in \mathcal{S}_{18} : x_B = 11\}, \quad R = \{x \in \mathcal{S}_{18} : x_A = 00\}.$$

Direct section comparison gives

$$\mathcal{S}_{18} = L \cup R, \quad L \cap R = \{110000, 110100, 111000, 111100\} = 11**00.$$

Both L and R are eleven-vertex copies of $\mathcal{L}_4$, and the displayed intersection is a maximal square in each. The Hamming gate of a vertex is obtained by overwriting the fixed coordinates defining the corresponding piece; inspection of (10) shows that the overwritten vertex is present. Hence both pieces are gated. A direct connectivity check after deletion of each of the eighteen vertices shows that there is no cut vertex.

For the bounded uniqueness assertion, the expansion census takes all 806 ample dimension-five bases of order at most seventeen and enumerates every nonempty intersection and every component assignment outside it. Of 767 596 intersections and 741 781 assignments, 1 209 give periphery-free isometric section pairs. Exactly three labelled occurrences are ample, and all three canonicalize to (10). This proves the stated uniqueness. $\square$

9.6 A Hall-derived obstruction

The maximum VC-dimension-three class of Hall has 299 concepts and no corner [CCMW22, Theorem 4.5]. A short sequence of restrictions produces the following additional obstruction.

Proposition 9.12 (Hall-derived obstruction; computer-assisted). *The six-coordinate family*

$$\begin{aligned}\mathcal{H}_{22} = \{ & 000111, 001100, 001101, 001111, 010000, 010100, 010101, 010111, \\ & 011100, 011101, 011111, 101000, 101001, 101010, 101011, 101100, \\ & 101101, 101111, 110000, 110100, 111000, 111100\}\end{aligned}$$

is an ample, periphery-free partial cube and is pc-minor-minimal outside $\mathcal{R}_{\text{per}}$. In particular, $\mathcal{H}_{22} \in \mathcal{O}_{\text{per}}$.

Proof. The family is obtained from Hall's 299-concept class by the restrictions

$$r_{10}^1, r_4^0, r_1^0, r_0^0, r_7^1, r_2^0,$$

where indices count from the right and are renumbered from zero after each step. The resulting six-coordinate family is mapped to the displayed one by

$$(x_0, x_1, x_2, x_3, x_4, x_5) \mapsto (x_0, x_1, x_2, x_4, 1 - x_5, 1 - x_3).$$

These operations record the provenance of the example but are not used in its verification. Applied directly to the displayed family, the computation in [section 12](#) compares its shattering and strong-shattering complexes, tests both section containments in every coordinate, and checks every elementary proper pc-minor recursively. These three checks establish, respectively, ampleness, absence of a peripheral coordinate, and membership of every proper pc-minor in $\mathcal{R}_{\text{per}}$. $\square$

9.7 The smallest additional obstruction

The order of a partial cube is constrained by its expansion sites. The following elementary lemma closes the high-dimensional part of the small-order search.

Lemma 9.13 (Partial cubes of order at most $d + 2$). *Let G be a finite connected partial cube of isometric dimension d . Then $|V(G)| \geq d + 1$. Moreover:*

- (i) *if $|V(G)| = d + 1$, then G is a tree;*
- (ii) *if $|V(G)| = d + 2$, then G is unicyclic and its unique cycle is a square.*

In particular, every nontrivial partial cube with $|V(G)| \leq d + 2$ has a peripheral Θ -class.

Proof. Reverse a complete sequence of d Θ -class contractions from G to K_1 , and let s_i be the number of vertices in the site duplicated by the i th reverse expansion. The new class has one edge for each vertex of that site, and the expansion adds exactly s_i vertices. Hence

$$|V(G)| = 1 + \sum_{i=1}^d s_i, \quad |V(G)| - (d + 1) = \sum_{i=1}^d (s_i - 1). \quad (11)$$

Every s_i is positive, proving the lower bound.

If equality holds, every new Θ -class is a singleton. Such a class is a one-edge cut and hence a bridge, so reversing the expansions from K_1 produces a tree.

If $|V(G)| = d + 2$, exactly one duplicated site has order two and all the others have order one. Immediately before the exceptional expansion, the graph is a tree. The two cover members are isometric subtrees, and their two-vertex intersection is therefore an edge. Duplicating it creates one square and no other cycle. Each later singleton class is a bridge and cannot alter the unique cyclic block. Thus the final graph is a square with trees attached to its vertices.

A nontrivial tree has a leaf edge, hence a peripheral class. If trees are attached to the square, a leaf outside the square again gives one; the bare square has both coordinates peripheral. $\square$

Theorem 9.14 (Global vertex-minimality; computer-assisted). *The pentagonal graph $\mathcal{P}_5$ has the smallest possible isometric dimension among the members of $\mathcal{O}_{\text{per}} \setminus \{Q_n^- : n \geq 3\}$. It is the unique additional obstruction of isometric dimension five and order at most twenty-one. Moreover, no additional obstruction has fewer than sixteen vertices. Consequently $\mathcal{P}_5$ is vertex-minimal among all additional obstructions, without a dimension restriction.*

Proof. The exact coordinate-isomorphism enumeration through isometric dimension four contains no additional obstruction. In dimension five, the exhaustive expansion enumeration through order twenty-one produces 14 259 partial-cube representatives, of which 3 115 are periphery-free; the only additional minimal type is $\mathcal{P}_5$. This proves the first two assertions.

For orders below sixteen, the exact expansion searches in dimensions six through twelve produce no ample periphery-free partial cube. Every additional obstruction is ample by [proposition 6.5](#), so this is the required exclusion. The counts and the exhaustiveness contract are recorded in [section 12](#). If $13 \leq d \leq 14$ and $|V(G)| \leq 15$, then $|V(G)| \leq d + 2$, so [lemma 9.13](#) supplies a peripheral class. Finally, a connected partial cube of dimension at least fifteen has at least $d + 1 \geq 16$ vertices. These cases exhaust all dimensions. $\square$

10 Gated and one-vertex amalgams

One-vertex identifications also occur in Polat’s separation of convex-treelike and faithful-treelike graphs [[Pol12](#), pp. 69–70]; these classes are compared in [section 10.2](#). Here we characterize both membership and pc-minor minimality for one-vertex amalgams by conditions on the pointed factors.

A *gated amalgam* is a union $G = G_1 \cup G_2$ of two intersecting gated subgraphs. Call a finite partial cube *gated-prime* if it is not such a union with both pieces proper. Minimal obstructions give an exact reduction of the gated-amalgamation question.

Proposition 10.1 (Gated-amalgam reduction). *The class $\mathcal{R}_{\text{per}}$ is closed under gated amalgamation if and only if every member of $\mathcal{O}_{\text{per}}$ is gated-prime. Every Q_n^- , $n \geq 3$, is gated-prime.*

Computer-assisted additional assertions. *The examples $\mathcal{P}_5$ and $\mathcal{H}_{22}$ are also gated-prime.*

Proof. Suppose $O \in \mathcal{O}_{\text{per}}$ is the gated amalgam of proper subgraphs A, B . Gated subgraphs are convex, hence A, B are proper pc-minors of O and belong to $\mathcal{R}_{\text{per}}$. Closure under gated amalgamation would incorrectly put O in the class.

Conversely, assume every member of $\mathcal{O}_{\text{per}}$ is gated-prime and let $G = G_1 \cup G_2$ be a gated amalgam with $G_1, G_2 \in \mathcal{R}_{\text{per}}$. If $G \notin \mathcal{R}_{\text{per}}$, take elementary pc-minors

$$H_0 = G, H_1, \dots, H_m = O$$

ending at $O \in \mathcal{O}_{\text{per}}$. Track two pieces A_k, B_k , beginning with $A_0 = G_1, B_0 = G_2$. Under restriction to a nonempty semicube W , put $A_{k+1} = A_k \cap W$ and $B_{k+1} = B_k \cap W$; under contraction π_e , take their images.

Nonempty restrictions of gated pieces remain gated: if $x, y \in W$ with y in the piece, then the gate of x lies in $I(x, y) \subseteq W$. Contraction images of gated subgraphs are gated by [[CKM20](#), Lemma 10]. At every stage the two images cover the current graph and are pc-minors of G_1, G_2 , so they belong to $\mathcal{R}_{\text{per}}$. Neither image can be empty or the whole current graph, since then the current graph, and hence its later minor O , would belong to $\mathcal{R}_{\text{per}}$. They continue to intersect: this is immediate under contraction, while after a restriction disjointness would disconnect the current semicube because no edge joins the two exclusive parts. Thus the final images form a proper gated decomposition of O , contradicting gated-primality.

For Q_n^{--} , every convex subgraph is the intersection with a Boolean face. If one of two proper faces in a putative intersecting cover has codimension at least two, their union has at most

$$2^{n-1} + 2^{n-2} - 1 < 2^n - 2$$

vertices. Otherwise both are halfcubes. Halfcubes in the same coordinate are equal or disjoint; in distinct coordinates their union has at most $3 \cdot 2^{n-2} - 1 < 2^n - 2$ vertices. Hence they cannot cover Q_n^{--} . For $\mathcal{P}_5$ and $\mathcal{H}_{22}$, the exact coordinate-face audit described in [section 12](#) tests all 3^5 , respectively 3^6 , intersections of semicubes, retains the gated ones, and finds no proper intersecting two-cover. $\square$

The last conclusion does not extend to all obstructions. To see the exact failure, we refine the recursion at a distinguished vertex.

Definition 10.2 (Root-avoiding decompositions and critical roots). A *root-avoiding peripheral decomposition* of a pointed partial cube (A, a) is defined recursively. A singleton has such a decomposition. At a nontrivial node, the chosen peripheral semicube must omit a ; its duplicated site has an ordinary recursive peripheral decomposition, while the contraction base, rooted at the image of a , again has a root-avoiding decomposition. Let $\mathcal{R}_{\text{per}}^\bullet$ be the class of pointed partial cubes admitting such a decomposition.

A nontrivial pointed member (A, a) of $\mathcal{R}_{\text{per}}$ is *root-locked* if every peripheral semicube contains a . It is *root-critical* if it is root-locked and, for every coordinate e , both the restriction of A to the e -semicube containing a and the contraction A/e , with their natural roots, belong to $\mathcal{R}_{\text{per}}^\bullet$.

For pointed graphs $(A, a), (B, b)$ on disjoint coordinate sets, write $A \vee B$ for their one-vertex amalgam obtained by identifying a and b . Distances between different factors add through the common vertex, so this amalgam is again a partial cube.

Example 10.3 (Why the root matters). Root the path $\{00, 10, 11\}$ at 00. First remove the peripheral side $\{11\}$, then the remaining endpoint different from the root. This is a root-avoiding decomposition. If another partial cube is attached at 00, these same two sides still omit the attachment vertex and can be removed in the same order. In contrast, attaching a nontrivial graph to the removed endpoint would enlarge that side and can destroy its peripherality. The wedge theorem formalizes this distinction; the eleven-vertex example in [proposition 10.7](#) will show that a member of $\mathcal{R}_{\text{per}}$ need not admit root avoidance at every vertex.

Lemma 10.4 (Peeling one factor inside a wedge). *If $(A, a) \in \mathcal{R}_{\text{per}}^\bullet$ and $B \in \mathcal{R}_{\text{per}}$, then the one-vertex amalgam obtained by identifying a with any vertex of B belongs to $\mathcal{R}_{\text{per}}$.*

Proof. Induct on $\text{idim}(A)$. For a singleton, the amalgam is B . Otherwise choose the first coordinate in the rooted decomposition. Its peripheral semicube omits a , so adjoining B at a neither changes that semicube nor spoils its peripherality. The contraction is the one-vertex amalgam of B with the rooted contraction base of A , and the duplicated site is unchanged. The induction hypothesis puts the former in $\mathcal{R}_{\text{per}}$, while the decomposition puts the latter there. The inverse peripheral expansion proves the claim. $\square$

Theorem 10.5 (Rooted-wedge characterization). *Let (A, a) and (B, b) be pointed members of $\mathcal{R}_{\text{per}}$.*

(i)

$$A \vee B \in \mathcal{R}_{\text{per}} \iff (A, a) \in \mathcal{R}_{\text{per}}^\bullet \text{ or } (B, b) \in \mathcal{R}_{\text{per}}^\bullet.$$

- (ii) If A, B are nontrivial, then $A \vee B$ is periphery-free if and only if both pointed factors are root-locked.
- (iii) If A, B are nontrivial, then

$$A \vee B \in \mathcal{O}_{\text{per}} \iff (A, a) \text{ and } (B, b) \text{ are root-critical.}$$

Moreover, $\mathcal{R}_{\text{per}}^\bullet$ is closed under contractions and restrictions to semicubes containing the root. Consequently, the root-critical graphs are exactly the pointed-minor-minimal members of

$$\{(A, a) : A \in \mathcal{R}_{\text{per}}\} \setminus \mathcal{R}_{\text{per}}^\bullet,$$

where pointed minors use those two root-preserving operations.

The proof uses the same commutation argument as ordinary heredity, with the additional requirement that the removed side avoid the root. In a wedge, a coordinate belongs to one factor: the other factor enlarges only its root side. This identifies the possible peripheral steps and reduces both membership and minimality to the pointed factors.

Proof. We first prove closure under root-preserving minors. Commute one such operation in coordinate f past the first coordinate e of a root-avoiding decomposition. If $f = e$, either contraction or restriction to the root side leaves the rooted contraction base. If $f \neq e$, the new contraction base and duplicated site are the corresponding f -minors of the old ones, and the chosen peripheral side still omits the root. If that side becomes empty, or if e becomes constant, the result is a root-preserving minor of the old contraction base. Induction on isometric dimension, using [lemma 5.1](#) for the unpointed duplicated site, proves the asserted closure.

For (i), the reverse implication is [lemma 10.4](#) and its symmetric version. For necessity, induct on $\text{idim}(A) + \text{idim}(B)$. A singleton factor already lies in $\mathcal{R}_{\text{per}}^\bullet$, so assume both are nontrivial and use disjoint coordinate sets. For a coordinate e of A , write A_e^a and $A_e^{\bar{a}}$ for the projected sections, with the first containing a . The corresponding sections of the wedge are

$$A_e^a \vee B \quad \text{and} \quad A_e^{\bar{a}}.$$

The first contains a vertex of the nontrivial factor B outside the A -coordinate block, so it cannot be contained in the second. Consequently, e is peripheral in the wedge exactly when $A_e^{\bar{a}} \subseteq A_e^a$; in that case the peripheral side omits the root.

Choose the first coordinate of a recursive decomposition of $A \vee B$, say it lies in A . Its contraction base is $A_e^a \vee B$ and belongs to $\mathcal{R}_{\text{per}}$. By induction, either (A_e^a, a) or (B, b) lies in $\mathcal{R}_{\text{per}}^\bullet$. In the second case we are done. In the first, $A_e^{\bar{a}}$ is a pc-minor of A and hence lies in $\mathcal{R}_{\text{per}}$. Prepending the peripheral coordinate e gives a root-avoiding decomposition for (A, a) . The case in which the first coordinate lies in B is symmetric.

The same section calculation proves (ii): a wedge coordinate is peripheral exactly when its factor has a peripheral semicube avoiding the root.

Suppose both factors are root-critical. They do not lie in $\mathcal{R}_{\text{per}}^\bullet$, so (i) puts their wedge outside $\mathcal{R}_{\text{per}}$. For a coordinate e of A , restriction to the side omitting the root deletes B and gives a pc-minor of A . Restriction to the root side and contraction give

$$A_e^a \vee B \quad \text{and} \quad (A/e) \vee B.$$

Both lie in $\mathcal{R}_{\text{per}}$ by (i) and root-criticality. The same holds for coordinates of B , so every elementary proper pc-minor of the wedge is in the class. Hence $A \vee B \in \mathcal{O}_{\text{per}}$.

Conversely, let $A \vee B \in \mathcal{O}_{\text{per}}$. Every obstruction is periphery-free: otherwise its peripheral restriction and contraction would be proper minors in $\mathcal{R}_{\text{per}}$ and would reconstruct it by a permitted recursive expansion. Part (ii) therefore makes both factors root-locked. For each coordinate e of A , the two displayed root-preserving minors of the wedge lie in $\mathcal{R}_{\text{per}}$. The factor (B, b) is root-locked and hence does not lie in $\mathcal{R}_{\text{per}}^\bullet$; part (i) forces (A_e^a, a) and $(A/e, \pi_e(a))$ into $\mathcal{R}_{\text{per}}^\bullet$. Thus (A, a) is root-critical, and symmetry gives the same for (B, b) . This proves (iii).

Finally, a root-critical graph lies outside $\mathcal{R}_{\text{per}}^\bullet$ and all its immediate root-preserving minors lie inside. Conversely, a pointed-minimal nonmember cannot have a peripheral semicube avoiding the root: its contraction base would be a proper root-preserving pointed minor with a root-avoiding decomposition, and prepending that coordinate would give one for the original graph. It is therefore root-locked, and pointed minimality supplies the remaining criticality conditions. $\square$

Corollary 10.6 (Root avoidance through convex subgraphs). *Let $A \in \mathcal{R}_{\text{per}}$ and $a \in V(A)$. Then $(A, a) \in \mathcal{R}_{\text{per}}^\bullet$ if and only if every nontrivial convex subgraph of A containing a has a peripheral semicube that omits a .*

Proof. A convex subgraph containing a is an intersection of semicubes containing a . Thus the forward implication follows from the pointed minor closure in [theorem 10.5](#). Conversely, apply the stated condition to a nontrivial A and choose a peripheral semicube avoiding a . Its opposite semicube is the rooted contraction base. Every convex subgraph of this base is convex in A , so the base inherits the condition and has a root-avoiding decomposition by induction on dimension. The duplicated site is a convex semicube of A and hence belongs to $\mathcal{R}_{\text{per}}$. These two decompositions give the required rooted step. A singleton starts the induction. $\square$

This is a pointed version of Polat's convex-subgraph condition: the additional requirement is that each peripheral side avoid the specified vertex. It supplies a geometric test for the recursive rooted property.

10.1 A rooted-wedge obstruction

Let

$$\mathcal{L}_4 = \{0000, 0100, 0101, 0111, 1000, 1010, 1011, 1100, 1101, 1110, 1111\} \subseteq Q_4 \quad (12)$$

with distinguished root 0^4 , and put

$$\mathcal{W}_8 = (\mathcal{L}_4 \times \{0^4\}) \cup (\{0^4\} \times \mathcal{L}_4) \subseteq Q_8. \quad (13)$$

The two factors meet only at 0^8 .

Proposition 10.7 (Rooted-wedge obstruction). *The graph $\mathcal{L}_4$ belongs to $\mathcal{R}_{\text{per}}$, but every peripheral semicube of $\mathcal{L}_4$ contains its root. The graph $\mathcal{W}_8$ is an ample VC-dimension-two partial cube of order twenty-one and*

$$\mathcal{W}_8 \in \mathcal{O}_{\text{per}}.$$

Its crossing graph is $K_4 \dot{\cup} K_4$, and it is the gated amalgam of two proper members of $\mathcal{R}_{\text{per}}$ along one vertex. Consequently:

- (i) $\mathcal{R}_{\text{per}}$ is not closed under gated amalgamation;
- (ii) not every member of $\mathcal{O}_{\text{per}}$ is gated-prime;
- (iii) an additional minimal obstruction need not shatter every coordinate pair.

Proof. A direct trace calculation gives

$$\text{str}(\mathcal{L}_4) = \{S \subseteq [4] : |S| \leq 2\}.$$

Thus $\mathcal{L}_4$ is maximum of VC dimension two, hence ample. It is cube-isomorphic to H_{11} in the proof of [proposition 9.1](#), whose displayed decomposition puts it in $\mathcal{R}_{\text{per}}$. Comparing its two sections in each coordinate shows that its only peripheral semicubes are $\{x_1 = 0\}$ and $\{x_2 = 0\}$. Both contain 0^4 , so $(\mathcal{L}_4, 0^4) \notin \mathcal{R}_{\text{per}}^\bullet$.

We verify the root-preserving minors needed for criticality. Suppress the fixed coordinate and keep the order of the other three. The root-side restrictions and contractions are

i	$\mathcal{L}_{4,i}^0$	$\pi_i \mathcal{L}_4$
1	$\{000, 100, 101, 111\}$	$\{000, 010, 011, 100, 101, 110, 111\}$
2	$\{000, 100, 110, 111\}$	$\{000, 001, 011, 100, 101, 110, 111\}$
3	$\{000, 010, 011, 100, 110, 111\}$	$\{000, 010, 011, 100, 101, 110, 111\}$
4	$\{000, 010, 100, 101, 110, 111\}$	$\{000, 010, 011, 100, 101, 110, 111\}$.

[Appendix A](#) gives root-avoiding decompositions of both entries in each row. Hence every root-side restriction and contraction has a root-avoiding decomposition. Together with the preceding periphery calculation, this proves that $(\mathcal{L}_4, 0^4)$ is root-critical. The opposite-side restrictions are proper pc-minors of $\mathcal{L}_4$ and belong to $\mathcal{R}_{\text{per}}$ by [lemma 5.1](#).

The one-vertex union $\mathcal{W}_8$ is isometric: within a factor this is true for $\mathcal{L}_4$, while every shortest path between distinct factors passes through 0^8 and has the sum of the two Hamming lengths. Its shattered and strongly shattered sets are the two copies of those of $\mathcal{L}_4$, with the empty set counted once. They consist of the empty set, the eight singletons, and the twelve pairs lying within one coordinate block. This proves ampleness, VC dimension two, order $1 + 8 + 12 = 21$, and crossing graph $K_4 \dot{\cup} K_4$.

Since both factors are root-critical, [theorem 10.5\(iii\)](#) gives $\mathcal{W}_8 \in \mathcal{O}_{\text{per}}$. Finally, each copy of $\mathcal{L}_4$ is gated in the wedge: a vertex in the other factor has the common root as its unique gate. Thus $\mathcal{W}_8$ is a proper gated amalgam of two members of $\mathcal{R}_{\text{per}}$, proving all three consequences. $\square$

10.2 Polat's pointed construction and its forbidden minor

An isometric subgraph H of a partial cube G is *faithful* if it is median-stable: whenever three vertices of H have a median in G , that median belongs to H [[Pol12](#), p. 66]. Here a median is a vertex belonging to all three pairwise intervals; in an isometric cube embedding it must be the coordinatewise majority vector. Every convex subgraph is faithful. Polat calls G *faithful-treelike* when every nontrivial finite faithful subgraph has a periphery [[Pol12](#), Definition 5.5]. This condition implies convex-treelikeness. His example on pp. 69–70 separates the two conditions by joining two copies of a cube with five vertices removed from one layer. The following construction makes the role of the pointed factors explicit.

Proposition 10.8 (Extending Polat's pointed construction). *Let $(A, 0^d)$ and $(B, 0^e)$ be nontrivial root-locked members of $\mathcal{R}_{\text{per}}$, with irredundant coordinate embeddings. Put*

$$P(A) = (Q_d \times \{0\}) \cup (A \times \{1\}), \quad P(B) = (Q_e \times \{0\}) \cup (B \times \{1\}),$$

and identify their bottom origins to form $G = P(A) \vee P(B)$, with common vertex b . Let F be the induced subgraph consisting of the two top layers and b . Then G is convex-treelike, whereas F is faithful and periphery-free, with convex hull G . In F , contracting the two edges joining the top origins to b gives $A \vee B$. This quotient is in $\mathcal{O}_{\text{per}}$ if and only if both pointed factors are root-critical.

Proof. The top layer of $P(A)$ is a peripheral side avoiding its bottom origin. Removing it leaves Q_d , which has a root-avoiding decomposition at every vertex, while its duplicated site is $A \in \mathcal{R}_{\text{per}}$. Thus $P(A)$ has a root-avoiding decomposition, as does $P(B)$. [Theorem 10.5\(i\)](#) puts G in $\mathcal{R}_{\text{per}}$.

Use disjoint coordinate blocks for the two copies. Within each top layer, distances in F agree with Hamming distances; between the layers, shortest paths pass through their origins and b , again attaining the Hamming distance. Thus F is isometric. If two vertices of a triple lie in one top layer, their majority vector has that layer coordinate equal to one. If this vector belongs to G , it therefore belongs to the same top layer, hence to F . A triple with no two vertices in one top layer consists of one vertex from each and b ; its majority is b . Triples with repeated vertices need no further check. This proves median stability. Every coordinate varies on F , so no proper semicube of G contains it. Its convex hull is therefore G .

For an old coordinate of A , the side avoiding its root is not peripheral by root-locking; the vertices witnessing this remain in F . The root side contains the other top layer, whose vertices have no neighbors across that coordinate. Thus neither side is peripheral. The same argument applies to B . Each new layer coordinate has just one edge across it, and both sides contain more than one vertex; neither side is peripheral. Finally, contracting these two single-edge coordinate classes identifies the two top origins with b , giving $A \vee B$. Its minimality criterion is [theorem 10.5\(iii\)](#). $\square$

For $A = B = \mathcal{L}_4$, the orders in this construction are

$$\underbrace{G}_{53 \text{ vertices}} \supsetneq \underbrace{F}_{23 \text{ vertices}} \xrightarrow{\text{two coordinate contractions}} \underbrace{\mathcal{W}_8}_{21 \text{ vertices}}.$$

The first inclusion is faithful, not convex; in particular it is not a conditioning operation. Polat's text specifies an eleven-vertex rooted convex-treelike factor in Q_4 , all of whose peripheral sides contain the root. An exact check of all $\binom{15}{10} = 3003$ eleven-vertex subsets containing 0^4 finds a unique pointed type with these properties, namely $(\mathcal{L}_4, 0^4)$. Thus the identification with his factor is computer-assisted and uses the properties stated in his text. The verification does not depend on a transcription of Figure 3. The factor and the gluing construction are already present in Polat's example; the rooted-wedge criterion identifies precisely when the contracted graph is a minimal forbidden pc-minor.

Corollary 10.9 (Classification of one-vertex-decomposable obstructions). *The members of $\mathcal{O}_{\text{per}}$ admitting a decomposition into two nontrivial partial cubes meeting in one vertex are exactly the wedges of two root-critical pointed members of $\mathcal{R}_{\text{per}}$.*

Proof. In such a one-vertex decomposition, each factor is a proper convex subgraph and therefore a proper pc-minor of the obstruction; hence both factors belong to $\mathcal{R}_{\text{per}}$. The claim is then exactly [theorem 10.5\(iii\)](#). $\square$

Proposition 10.10 (Root-critical wedges admit corner peelings). *Every root-critical pointed member (A, a) of $\mathcal{R}_{\text{per}}$ has a corner peeling ending at a . Consequently every one-vertex-decomposable member of $\mathcal{O}_{\text{per}}$ is corner-peelable.*

Proof. It is convenient to use the equivalent reverse formulation: a corner peeling ending at a is an ordering beginning at a whose every prefix induces an isometric subgraph.

We first claim that every $(A, a) \in \mathcal{R}_{\text{per}}^\bullet$ has such an ordering. Induct through a root-avoiding peripheral decomposition and orient its first expansion as

$$A = (B \times \{0\}) \cup (S \times \{1\}), \quad S \subseteq B,$$

with root $(a, 0)$. By induction, order the B -layer from a , and then append any isometric-prefix ordering of S . A prefix in the second block is

$$(B \times \{0\}) \cup (T \times \{1\}),$$

where T is isometric in S , and hence in B . Cross-layer distances are realized by a B -geodesic ending in T , followed by the new-coordinate edge. Thus every prefix is isometric.

Now let (A, a) be root-critical and choose a peripheral coordinate. Root-locking orients its expansion as

$$A = (S \times \{0\}) \cup (B \times \{1\}), \quad a \in S \subseteq B.$$

Root-criticality puts the root-side restriction (S, a) and the contraction (B, a) in $\mathcal{R}_{\text{per}}^\bullet$. If

$$a = b_1, b_2, \dots, b_p \quad \text{and} \quad a = s_1, s_2, \dots, s_q$$

are the corresponding isometric-prefix orderings, then

$$(a, 0), (a, 1), (b_2, 1), \dots, (b_p, 1), (s_2, 0), \dots, (s_q, 0)$$

is an isometric-prefix ordering of A . Before the B -layer is complete, every cross-layer distance passes through the edge $(a, 0)(a, 1)$; afterwards, a cross-layer geodesic from $(s, 0)$ crosses at $(s, 1)$ and continues inside the complete B -layer.

Finally, by [corollary 10.9](#), a one-vertex-decomposable obstruction is a wedge $A \vee B$ of two root-critical pointed factors. Take their isometric-prefix orderings, list the common root first, then the remaining vertices of A , and then those of B . Distances between the factors add through the root, so every prefix is isometric. Reversing the ordering gives a corner peeling of the wedge. $\square$

Remark 10.11 (Small-dimensional pointed enumeration; computer-assisted). Through isometric dimension four there are 233 pointed partial-cube types and 187 pointed types whose underlying graph belongs to $\mathcal{R}_{\text{per}}$. Exactly one of the latter is root-critical: $(\mathcal{L}_4, 0^4)$, up to pointed cube isomorphism. The exact program checks [theorem 10.5\(i\)](#) and its minimality criterion on all 17 578 unordered pairs with repetition of those 187 types.

The dimension-five expansion enumeration through order sixteen contains 679 ordinary and 7 362 pointed $\mathcal{R}_{\text{per}}$ -types. Exactly fourteen pointed types are root-critical: two have order fourteen, five have order fifteen, and seven have order sixteen. Wedging each with $(\mathcal{L}_4, 0^4)$ yields further dimension-nine excluded pc-minors of orders twenty-four, twenty-five, and twenty-six. The computation records all fourteen vertex lists and independently checks every elementary pc-minor of the resulting wedges.

11 Two-layer recognition and repair antichains

The preceding structural results do not require inspecting all minors. For a general candidate obstruction, however, the coordinate criterion in [theorem 6.1](#) is an exact recognition procedure. We now express that procedure using the two sections at a chosen coordinate and track how their failed containments change under projection and conditioning.

11.1 Two layers and failed section containments

Let $X \subseteq \{0, 1\}^E$, let $p \in E$, and let $b \in \{0, 1\}$. Define the set of witnesses to failure of the containment $X_p^{1-b} \subseteq X_p^b$ by

$$\mathcal{D}_X(p, b) = X_p^{1-b} \setminus X_p^b. \tag{14}$$

Thus p , oriented with base side $p = b$, is peripheral in X exactly when $\mathcal{D}_X(p, b) = \emptyset$.

For two families $A, B \subseteq \{0, 1\}^E$, put

$$\Lambda(A, B) = (A \times \{0\}) \cup (B \times \{1\}). \quad (15)$$

We assume that this is an irredundant partial-cube representation. The new coordinate is called the *outer coordinate*; the coordinates in E are the *layer coordinates*.

Lemma 11.1 (Peripheral coordinates of a two-layer partial cube). *The partial cube $\Lambda(A, B)$ has no peripheral coordinate if and only if*

$$A \not\subseteq B, \quad B \not\subseteq A, \quad (16)$$

and

$$\mathcal{D}_A(p, b) \cup \mathcal{D}_B(p, b) \neq \emptyset \quad (p \in E, b \in \{0, 1\}). \quad (17)$$

Proof. The outer coordinate is peripheral in one orientation exactly when $A \subseteq B$ or $B \subseteq A$. For a layer coordinate p , its $p = 1 - b$ section is contained in its $p = b$ section precisely when this containment holds in both layers. By (14), this is equivalent to $\mathcal{D}_A(p, b) \cup \mathcal{D}_B(p, b) = \emptyset$. These are all coordinates and both orientations. $\square$

A *simultaneous descendant* of (A, B) is obtained by applying the same finite sequence of layer-coordinate restrictions and contractions to both families. Empty layers are allowed during this process. After each operation, coordinates constant in the complete two-layer family are deleted. A descendant is *proper* when the operation sequence is nonempty.

Example 11.2 (The six-cycle). Let $E = \{1, 2\}$,

$$A = 2^E \setminus \{\emptyset\}, \quad B = 2^E \setminus \{E\}.$$

Then $\Lambda(A, B) \cong Q_3^{--} = C_6$. The layers are incomparable. If q is the unique coordinate in $E \setminus \{p\}$, direct calculation gives

$$\mathcal{D}_A(p, 0) = \{\emptyset\}, \quad \mathcal{D}_B(p, 1) = \{\{q\}\}.$$

Thus A witnesses the orientation $b = 0$, whereas B witnesses the orientation $b = 1$, for each $p \in E$; hence (17) holds. This illustrates the coverage condition in lemma 11.1. The lift is not ample, so it belongs to the nonample family in proposition 6.4.

Theorem 11.3 (Complete two-layer forbidden-minor basis). *Let $O = \Lambda(A, B)$ be an irredundant ample partial cube. Then $O \in \mathcal{O}_{\text{per}}$ if and only if:*

- (i) $A, B, A \cup B \in \mathcal{R}_{\text{per}}$;
- (ii) (16) and (17) hold; and
- (iii) no proper simultaneous descendant satisfies both conditions in (ii) on its surviving coordinates.

Thus, up to partial-cube isomorphism,

$$\mathcal{O}_{\text{per}} = \{Q_n^{--} : n \geq 3\} \dot{\cup} \{\Lambda(A, B) : \Lambda(A, B) \text{ is ample and (i)–(iii) hold}\}. \quad (18)$$

Every coordinate of an ample obstruction may be chosen as the outer coordinate in this description.

Proof. By proposition 6.4, every Q_n^{--} , $n \geq 3$, belongs to $\mathcal{O}_{\text{per}}$. Conversely, if $O \in \mathcal{O}_{\text{per}}$ is not ample, the forbidden-minor characterization of ample partial cubes [KM20, Corollary 7.4] gives a pc-minor of O isomorphic to some Q_n^{--} . This minor lies outside $\mathcal{R}_{\text{per}}$, so the minimality of O forces $O \cong Q_n^{--}$.

Now let $O \in \mathcal{O}_{\text{per}}$ be ample, choose any coordinate e , and write

$$O = \Lambda(A, B), \quad A = O_e^0, \quad B = O_e^1.$$

The two restrictions A, B and the contraction $A \cup B$ are proper pc-minors of O , hence belong to $\mathcal{R}_{\text{per}}$. Every obstruction is periphery-free: if e were peripheral, its contraction base and duplicated site would be proper pc-minors in $\mathcal{R}_{\text{per}}$, and their permitted peripheral expansion would reconstruct O . Therefore lemma 11.1 gives (16)–(17). A proper simultaneous descendant gives a proper pc-minor of O , which belongs to $\mathcal{R}_{\text{per}}$ and consequently has a peripheral coordinate unless it is a singleton. In either case it fails the two conditions in lemma 11.1. Thus (A, B) satisfies (i)–(iii).

Conversely, suppose the ample partial cube $O = \Lambda(A, B)$ satisfies (i)–(iii). By lemma 11.1, O is periphery-free, so $O \notin \mathcal{R}_{\text{per}}$. We prove that every proper pc-minor of O belongs to $\mathcal{R}_{\text{per}}$.

First consider a minor operation sequence that uses the outer coordinate. Operations in distinct coordinates commute, so its first outer-coordinate operation may be moved to the beginning. The resulting graph is a pc-minor of A , B , or $A \cup B$, and therefore belongs to $\mathcal{R}_{\text{per}}$.

It remains to consider descendants retaining the outer coordinate. We induct on the number of surviving layer coordinates. Let $N = \Lambda(C, D)$ be a proper simultaneous descendant. If one layer is empty, the irredundant graph N is a pc-minor of A or B . If the layers are nested, the outer coordinate is peripheral. Its contraction base $C \cup D$ is obtained by applying the same layer operations to $A \cup B$, so it is a pc-minor of $A \cup B$. The duplicated site is the smaller of C, D , hence a pc-minor of A or B , respectively. Condition (i) and pc-minor closure put both children in $\mathcal{R}_{\text{per}}$. Their peripheral expansion is N , so again $N \in \mathcal{R}_{\text{per}}$.

Suppose instead that C, D are nonempty and incomparable. Condition (iii) gives a surviving layer coordinate p and an orientation b for which

$$\mathcal{D}_C(p, b) \cup \mathcal{D}_D(p, b) = \emptyset.$$

The coordinate p is therefore peripheral in N . Its contraction base and duplicated site are simultaneous descendants with one fewer layer coordinate, so the induction hypothesis puts both in $\mathcal{R}_{\text{per}}$. Their peripheral expansion is N . This proves that every proper pc-minor of O belongs to $\mathcal{R}_{\text{per}}$, and hence $O \in \mathcal{O}_{\text{per}}$. $\square$

11.2 Minimal coordinate sets that repair a containment

The preceding theorem uses descendants to express minimality. We next evaluate that condition using finite data computed from the two original layers. The data record how projection repairs a failed containment and how conditioning removes its witnesses. The test still ranges over coordinate states, but does not need to construct each resulting minor.

Let $O = \Lambda(A, B) \subseteq \{0, 1\}^{E \cup \{e\}}$, where e is the outer coordinate. For $p \in E \cup \{e\}$, $b \in \{0, 1\}$, and $u \in \mathcal{D}_O(p, b)$, define

$$\mathcal{H}_O(p, b; u) = \min_{\subseteq} \{\Delta(u, v) : v \in O_p^b\}, \quad \Delta(u, v) = \{q \neq p : u_q \neq v_q\}. \quad (19)$$

Thus $\mathcal{H}_O(p, b; u)$ is the antichain of the inclusion-minimal coordinate sets on which u differs from a possible repair in the base section. It is empty when O_p^b is empty.

A *minor state* is a map

$$\sigma : E \longrightarrow \{\mathbf{K}, \mathbf{C}, 0, 1\}.$$

The four symbols mean retained, contracted, restricted to zero, and restricted to one. Put

$$K_\sigma = \sigma^{-1}(\mathbf{K}), \quad C_\sigma = \sigma^{-1}(\mathbf{C}), \quad F_\sigma = \sigma^{-1}(\{0, 1\}).$$

For $p = e$, or for $p \in K_\sigma$, say that a defect $u \in \mathcal{D}_O(p, b)$ *survives* σ if

$$u_q = \sigma(q) \quad (q \in F_\sigma), \quad R \not\subseteq C_\sigma \quad (R \in \mathcal{H}_O(p, b; u)). \quad (20)$$

The first condition says that the defect survives the fixed coordinates. The second says that contraction has not supplied a repair: equivalently, C_σ is an independent set of the repair hypergraph $\mathcal{H}_O(p, b; u)$.

Let $\Sigma_e(O)$ be the set of minor states σ such that

- (a) for each $b \in \{0, 1\}$, the outer orientation (e, b) has a surviving defect; and
- (b) for each $p \in K_\sigma$ and $b \in \{0, 1\}$, the orientation (p, b) has a surviving defect.

Write $\mathbf{K}$ for the state retaining every coordinate. The recognition criterion will require that this set have a unique member:

$$\Sigma_e(O) = \{\mathbf{K}\}. \quad (21)$$

The condition is computed from the sections of O . It means that every proper coordinate operation destroys at least one of the required noncontainments.

Example 11.4 (Repairing a containment in the six-cycle). Use the layers $A = \{01, 10, 11\}$, $B = \{00, 01, 10\}$ of the earlier six-cycle example, with outer coordinate e . For the containment $A \subseteq B$, the only witness is $u = 11$. Its differences from the three possible mates in B have supports $\{1, 2\}$, $\{1\}$, $\{2\}$, so

$$\mathcal{H}_O(e, 1; 11) = \{\{1\}, \{2\}\}.$$

Projecting either layer coordinate supplies a mate for this witness. Fixing a layer coordinate to zero removes the witness; fixing it to one preserves it, provided the other coordinate is retained. The following three states show the different effects (the remaining bits are in coordinate order):

$\sigma(1), \sigma(2)$	$A[\sigma]$	$B[\sigma]$	$A[\sigma] \subseteq B[\sigma]$
$\mathbf{K}, \mathbf{K}$	$\{01, 10, 11\}$	$\{00, 01, 10\}$	no
$\mathbf{C}, \mathbf{K}$	$\{0, 1\}$	$\{0, 1\}$	yes
$1, \mathbf{K}$	$\{0, 1\}$	$\{0\}$	no

Here $A[\sigma]$ and $B[\sigma]$ mean applying the indicated layer operations to each family. In the last row only one outer containment fails: the reverse containment holds. Nonconstancy of the outer coordinate alone therefore does not imply failure in both orientations.

Lemma 11.5 (Static repair formula). *For a minor state σ , define the one-step family*

$$O[\sigma] = \{x|_{\{e\} \cup K_\sigma} : x \in O, x_q = \sigma(q) \text{ for } q \in F_\sigma\}. \quad (22)$$

Then $\sigma \in \Sigma_e(O)$ if and only if every displayed coordinate of $O[\sigma]$ has both orientations nonperipheral. Consequently, a proper simultaneous descendant of (A, B) satisfies (16) and (17) on its surviving layer coordinates if and only if

$$\Sigma_e(O) \setminus \{\mathbf{K}\} \neq \emptyset.$$

Proof. Fix a displayed coordinate p and an orientation b . Suppose first that the $p = 1 - b$ section of $O[\sigma]$ contains a defect z . Choose a preimage $u \in O_p^{1-b}$ of z satisfying the restrictions. Since a copy of u in O_p^b would also survive and project to z , we have $u \in \mathcal{D}_O(p, b)$. If some $R \in \mathcal{H}_O(p, b; u)$ were contained in C_σ , a corresponding base vertex would agree with u on all retained and fixed coordinates; it would therefore project to z , a contradiction. Thus u survives σ .

Conversely, let $u \in \mathcal{D}_O(p, b)$ survive σ . It survives all fixed coordinates. If its projection had a mate in the $p = b$ section of $O[\sigma]$, some $v \in O_p^b$ would differ from u only on contracted coordinates. Hence $\Delta(u, v) \subseteq C_\sigma$, and an inclusion-minimal member of the family in (19) would also be contained in C_σ , contrary to (20). The projection of u is therefore a defect of the displayed orientation. Applying this equivalence to both orientations of e and of every coordinate in K_σ proves the first assertion.

Operations in distinct coordinates commute, and a coordinate disappears after its first operation. Hence every simultaneous descendant retaining e has the form (22). Conversely, whenever $O[\sigma]$ is nonempty, it is realized by the indicated restrictions and contractions; every prefix of the restrictions remains nonempty because it contains a preimage of a member of $O[\sigma]$. If a displayed coordinate becomes constant, contracting it deletes only redundant labelling. Repeating this deletion gives another state and the normalized descendant. The outer coordinate has surviving defects in both orientations precisely when the two resulting layers are incomparable; in particular, it is then nonconstant. If the normalized descendant has no peripheral coordinate, encode every deleted constant layer coordinate by $\mathbf{C}$ instead of $\mathbf{K}$. This changes only redundant coordinates and gives a state in $\Sigma_e(O)$ by the first assertion. Conversely, a state in $\Sigma_e(O)$ has no constant displayed coordinate and satisfies the two noncontainment conditions by lemma 11.1. A nonidentity state deletes at least one original layer coordinate, so the descendant is proper. This proves the final equivalence. $\square$

Theorem 11.6 (Recognition of ample obstructions by repair antichains). *Up to partial-cube isomorphism, the ample members of $\mathcal{O}_{\text{per}}$ are exactly the lifts $O = \Lambda(A, B)$ for which, at some outer coordinate e ,*

$$O \text{ is ample,} \quad A, B, A \cup B \in \mathcal{R}_{\text{per}}, \quad \Sigma_e(O) = \{\mathbf{K}\}. \quad (23)$$

Moreover, every coordinate of an ample member of $\mathcal{O}_{\text{per}}$ may be chosen as e . Together with the nonample family, this gives

$$\mathcal{O}_{\text{per}} = \{Q_n^{--} : n \geq 3\} \dot{\cup} \{\Lambda(A, B) : (23) \text{ holds}\}. \quad (24)$$

Proof. The all-retained state belongs to $\Sigma_e(O)$ exactly when O is periphery-free. By lemma 11.5, the unique-state condition says exactly that no proper simultaneous descendant of (A, B) retains

the noncontainment conditions. Thus, for an ample lift, the three conditions in (23) are equivalent to (i)–(iii) in [theorem 11.3](#). The claim follows from [theorem 11.3](#), including the assertion that every coordinate of an ample obstruction may serve as the outer coordinate. $\square$

Remark 11.7 (Scope of the recognition criteria). The coordinate criterion treats all coordinates symmetrically. The two-layer criterion singles out one coordinate, and the repair criterion expresses the effect of all remaining coordinate operations through the antichains (19). These descriptions recognize the same forbidden family. They do not classify the systems of antichains for which $\Sigma_e(O) = \{\mathbf{K}\}$. In particular, the direct repair test may inspect all $4^{|E|}$ states; no fixed bound on the size of a sufficient local test is asserted.

12 Computer-assisted finite classifications

The structural theorems above are symbolic. Computation is used for the finite uniqueness assertions for $\mathcal{P}_5$ and $\mathcal{S}_{18}$, the Hall-derived obstruction, gated-primality of displayed examples, and the finite pointed and small-order enumerations. This section makes the exhaustiveness contract explicit.

12.1 Representation and verification predicates

A coordinate family on d coordinates is stored as a bitset of the 2^d cube vertices. Cube isomorphism consists of permuting coordinates and independently complementing them. Canonical representatives are computed under this action. All metric, shattering, section-containment, restriction, contraction, gatedness, and recursive-membership predicates are exact.

For a family H , recursive membership in $\mathcal{R}_{\text{per}}$ tests every coordinate whose sections are nested. For each such coordinate it recursively tests the projection and the smaller section. Memoization is performed only after canonicalization; it changes running time, not the accepted family. Root-avoiding membership additionally requires that the smaller peripheral side omit the distinguished root.

Every irredundant d -dimensional partial cube is an expansion of a $(d-1)$ -dimensional contraction base B . Its two projected sections A_0, A_1 are isometric subgraphs covering B . If $I = A_0 \cap A_1$, no edge of $B - I$ joins the two exclusive parts, so every connected component of $B - I$ lies wholly on one side. The generator therefore enumerates:

1. every contraction base B up to cube isomorphism;
2. every nonempty candidate intersection $I \subseteq B$;
3. every assignment of the components of $B - I$ to the two sides;
4. the isometry, partial-cube, ampleness, periphery, and minor predicates for the resulting expansion.

This covers every expansion and is not a random search.

For the higher-dimensional order-fifteen search, only ample contraction bases are required: by [proposition 6.5](#), every additional obstruction is ample, and contractions preserve ampleness [[BCDK06](#), [KM20](#)]. A target of order at most fifteen has a base of order at most fourteen.

12.2 Enumeration results

The dimension-five audit expands all 48 dimension-four bases through order twenty-one. It examines 103 668 intersections and 92 927 side assignments, retaining 14 259 partial-cube representatives and 3 115 periphery-free representatives. The only additional minimal type is $\mathcal{P}_5$.

The dimension-six census through order eighteen expands all 806 ample dimension-five bases of order at most seventeen. It examines 767 596 intersections and 741 781 component assignments, retaining 1 209 periphery-free isometric section pairs. Exactly three labelled occurrences are ample, and they form the single cube-isomorphism type $\mathcal{S}_{18}$. Thus $\mathcal{S}_{18}$ is the unique ample periphery-free dimension-six type of order at most eighteen; its immediate minor certificates give [proposition 9.11](#). In particular, there is no ample periphery-free dimension-six class of order at most sixteen.

For dimension seven, the input consists of all 2 801 ample dimension-six types of order at most fourteen. Their order distribution from seven through fourteen is

$$11, 28, 47, 111, 196, 387, 712, 1309.$$

The search tests 409 471 intersections and 447 542 nonperipheral assignments. The periphery-free filter retains 5 522 assignments, of which 24 have two isometric sections. None of these lifts is ample.

For dimension eight, all 10 075 ample dimension-seven bases of order at most fourteen are retained. The search examines 1 018 758 intersections and 1 405 424 two-sided assignments. Exactly 4 713 covers have no peripheral coordinate, but none has two isometric sections, so none is a partial cube.

The remaining exact layers are:

d	bases	intersections	assignments	raw periphery-free	isometric pairs
9	21 623	1 653 371	2 896 920	2 799	0
10	31 332	1 861 676	4 095 427	903	0
11	32 471	1 486 833	3 938 951	141	0
12	24 361	814 316	2 380 554	8	0

Thus no ample periphery-free partial cube of order at most fifteen survives in dimensions six through twelve. [Lemma 9.13](#) closes dimensions thirteen and fourteen, and the general lower bound closes all larger dimensions. This proves the computational premise of [theorem 9.14](#).

12.3 Verification and reproducibility

The computational records separate fixed-example checks from exhaustive enumerations. The fixed-example supplement checks the four named ample obstructions and the printed decomposition expressions independently, using only the Python standard library. It also compares the repair formula with directly constructed minors for every coordinate state of those examples. These checks do not establish the census completeness, uniqueness, or global minimum claims; those require exhaustive enumeration.

Both supplements also contain a separate check of the comparison with Polat’s pointed example. Among the 3003 rooted eleven-vertex subsets of Q_4 , 451 are isometric and 56 have no peripheral side avoiding the root. These 56 form six pointed types, exactly one of which is convex-treelike. The report records a coordinate permutation to $\mathcal{L}_4$, verifies the 53- and 23-vertex constructions, and checks the two contractions to $\mathcal{W}_8$. It also tests [corollary 10.6](#) on the 407 convex-treelike labelled families in this domain. This finite check supports the source comparison; the general construction and rooted convex characterization have symbolic proofs.

A separate census supplement supplies a second implementation of the finite classifications. It starts by enumerating all cube subsets through dimension four, then generates expansion covers by assigning the components outside each overlap to the two sections. In higher dimensions it compares the generated types with the stored lists using isomorphism of their one-inclusion graphs. This comparison is valid because an irredundant isometric cube embedding is unique up to coordinate permutations and flips. The higher-dimensional comparison uses `nauty` [MP14] through `pynauty`, version 2.8.8.1. The supplement records the complete chain of ample contraction bases, the periphery-free exclusions through order fifteen, both bounded uniqueness checks, and the pointed counts. Its replay guide states the coverage argument, input hashes, software dependency, and exact commands.

The enumeration stores each cube-isomorphism class by a canonical bitset representative. The chunked enumerations record the range of input bases, the number and digest of the resulting representatives, and the digest of the source list. The merge procedure rejects gaps, overlaps, malformed counts, stale inputs, and digest mismatches.

Several comparisons provide independent checks. The tree-only part of the ample-base enumeration gives 106, 235, 551 types on ten, eleven, and twelve vertices, respectively. Separate routines verify the stated restriction sequence for the Hall example, every elementary minor of $\mathcal{P}_5, \mathcal{S}_{18}, \mathcal{H}_{22}, \mathcal{W}_8$, the gated coordinate faces, the rooted decompositions, and the pointed-factor enumerations. None of the symbolic results outside statements explicitly marked computer-assisted uses these calculations.

As an independent check of lemma 11.5, the root formula and direct minor construction agree on all 1 497 177 four-state assignments for irredundant partial cubes through dimension four, of which 1 493 248 occur in dimension four itself. For each of the 25 coordinate presentations of $\mathcal{P}_5, \mathcal{S}_{18}, \mathcal{H}_{22}, \mathcal{W}_8$, the all-retained state is the unique solution. These computations test the implementation of the formula; the proof of the lemma is symbolic.

13 Conclusion and further problem

The finite convex-treelike class introduced by Polat admits a peripheral recursion and is closed under pc-minors. It therefore has three equivalent but operationally different descriptions: binary expansion trees, a periphery in every nontrivial pc-minor, and a periphery in every nontrivial convex subgraph. It contains the incomparable median and daisy classes and is properly contained in the standard tree-like class. It also retains ampleness, corner-peelability, almost-median and proper-semi-median structure, and Cartesian products, but loses arbitrary isometric subgraphs and gated amalgamation.

The obstruction theory has several visible sources. Antipodally punctured cubes give an unavoidable infinite family, and the cyclic-arc classes $\mathcal{A}_n$ of theorem 9.4 give an infinite ample family extending $\mathcal{P}_5$. The step-two classes $\mathcal{F}_n$ of theorem 9.7 give a second one, of VC dimension three, whose smaller section defect in every coordinate has three vertices. The pentagonal and Hall-derived examples are gated-prime additional obstructions. The square amalgam $\mathcal{S}_{18}$ is the first displayed face-amalgam example without a cut vertex, while root-critical pointed factors generate exactly all one-vertex-decomposable obstructions. The global small-order enumeration shows that the additional phenomena begin with the sixteen-vertex pentagonal class.

The coordinate and repair criteria give exact recognition procedures for the full forbidden family. The rooted-wedge theorem gives a generative description for its one-vertex-decomposable members. A remaining question is to find comparably explicit constructions for the other ample obstructions, including those with no cut vertex. In terms of the repair criterion, this asks which systems of minimal repair antichains satisfy $\Sigma_e(O) = \{\mathbf{K}\}$, and whether that property can be recognized through a smaller structural description.

Open Question 13.1. Give an intrinsic generative description of the root-critical pointed factors and the ample obstructions with no cut vertex.

A Recursive decompositions for the finite examples

This appendix records the finite decompositions used in [propositions 9.1](#) and [10.7](#). Write $j(A; B)$ when coordinate j is peripheral, its contraction has recursive decomposition A , and its smaller section has decomposition B . Write $\bullet$ for K_1 , suppress constant coordinates after each step, and renumber the remaining coordinates from one, counting from the right in the displayed bitstrings. This is the integer-bit convention used by the decomposition expressions; the earlier table of the four root-side minors lists the deleted coordinates from left to right.

For the three minor types in the proof of [proposition 9.1](#), direct section comparison gives

$$\begin{aligned} H_6 : & \quad 3(1(1(\bullet; \bullet); \bullet); 1(\bullet; \bullet)); \bullet, \\ H_{10} : & \quad 3(1(1(\bullet; \bullet); 1(\bullet; \bullet)); 1(1(\bullet; \bullet); \bullet)); 1(1(\bullet; \bullet); \bullet)), \\ H_{11} : & \quad 3(1(1(\bullet; \bullet); 1(\bullet; \bullet)); 1(1(\bullet; \bullet); \bullet)); 1(1(1(\bullet; \bullet); \bullet); \bullet)). \end{aligned}$$

For pointed families, write $j^*(A; B)$ when the peripheral side in coordinate j omits the zero root, the contraction has rooted decomposition A , and the site has ordinary decomposition B . The five rooted decomposition types needed for the four rows in the proof of [proposition 10.7](#) are

$$\begin{aligned} \alpha &= 1^*(1^*(1^*(\bullet; \bullet); 1(\bullet; \bullet)); 1(\bullet; \bullet)), \\ \beta &= 1^*(1^*(1^*(\bullet; \bullet); \bullet); \bullet), \\ \gamma &= 2^*(1^*(1^*(\bullet; \bullet); \bullet); \bullet), \\ \delta &= 1^*(1^*(1^*(\bullet; \bullet); 1(\bullet; \bullet)); 1(1(\bullet; \bullet); \bullet)), \\ \varepsilon &= 2^*(1^*(1^*(\bullet; \bullet); 1(\bullet; \bullet)); 1(1(\bullet; \bullet); \bullet)). \end{aligned}$$

In the order displayed in that proof, the four (restriction, contraction) pairs have decomposition types

$$(\gamma, \delta), \quad (\beta, \varepsilon), \quad (\alpha, \delta), \quad (\alpha, \delta).$$

B Notation

Symbol	Meaning
Q_X, Q_n	hypercube on coordinate set X , or on n coordinates
Θ	Djoković–Winkler equivalence relation on edges
$\text{idim}(G)$	number of Θ -classes of a partial cube G
$W_x^\varepsilon, U_x^\varepsilon$	semicube and its boundary in coordinate x
$H_x, \pi_x H$	section and coordinate projection of a set family
$\text{str}(H), \text{sstr}(H)$	shattered and strongly shattered complexes
$G^\sharp$	crossing graph of the Θ -classes
$\mathcal{R}_{\text{per}}$	finite convex-treelike partial cubes, equivalently the peripheral recursion
$\mathcal{R}_{\text{per}}^\bullet$	pointed members with a root-avoiding decomposition
$\mathcal{O}_{\text{per}}$	pc-minor-minimal partial cubes outside $\mathcal{R}_{\text{per}}$
$O[\tau]$	direct four-state pc-minor determined by retaining, contracting, or fixing coordinates, equation (2)
$\text{var}(O[\tau])$	coordinates nonconstant in the state family $O[\tau]$
$\Lambda(A, B)$	two-layer partial cube with layers A and B , equation (15)
$\mathcal{D}_X(p, b)$	witnesses to the failure of the signed section containment $X_p^{1-b} \subseteq X_p^b$, equation (14)
$\mathcal{H}_O(p, b; u)$	antichain of minimal coordinate supports that repair the defect u , equation (19)
$K_\sigma, C_\sigma, F_\sigma$	coordinates retained, contracted, or fixed by a four-state assignment σ
$\Sigma_e(O)$	assignments preserving both orientations of every displayed coordinate; the obstruction test requires $\Sigma_e(O) = \{\mathbf{K}\}$
Q_n^-, Q_n^{--}	cube with one vertex, or one antipodal pair, deleted
$\mathcal{P}_5$	sixteen-vertex pentagonal obstruction, equation (5)
$\mathcal{S}_{18}$	eighteen-vertex square-amalgam obstruction, equation (10)
$\mathcal{H}_{22}$	twenty-two-vertex Hall-derived obstruction, proposition 9.12
$\mathcal{L}_4, \mathcal{W}_8$	root-critical factor and rooted-wedge obstruction, equations (12) and (13)

Assistance disclosure

OpenAI Codex was used during the development and revision of this manuscript in 2026, including proof development and review, computational verification, comparison with source papers, and expository revision. The author is responsible for the mathematical claims, citations, and final text.

References

- [BCDK06] Hans-Jürgen Bandelt, Victor Chepoi, Andreas W. M. Dress, and Jack H. Koolen. Combinatorics of lopsided sets. *Eur. J. Comb.*, 27(5):669–689, 2006.
- [BIK⁺02] Bostjan Bresar, Wilfried Imrich, Sandi Klavžar, Henry Martyn Mulder, and Riste Skrekovski. Tiled partial cubes. *Journal of Graph Theory*, 40(2):91–103, 2002.
- [BIK03a] Boštjan Brešar, Wilfried Imrich, and Sandi Klavžar. Fast recognition algorithms for classes of partial cubes. *Discrete Applied Mathematics*, 131(1):51–61, 2003.
- [BIK03b] Boštjan Brešar, Wilfried Imrich, and Sandi Klavžar. Tree-like isometric subgraphs of hypercubes. *Discussiones Mathematicae Graph Theory*, 23(2):227–240, 2003.
- [CCMW22] Jérémie Chalopin, Victor Chepoi, Shay Moran, and Manfred K. Warmuth. Unlabeled sample compression schemes and corner peelings for ample and maximum classes. *Journal of Computer and System Sciences*, 127:1–28, 2022.

- [CKM20] Victor Chepoi, Kolja Knauer, and Tilen Marc. Hypercellular graphs: partial cubes without Q_3^- as partial cube minor. *Discrete Mathematics*, 343:111678, 2020.
- [Gol14] Tanja Gologranc. Tree-like partial hamming graphs. *Discussiones Mathematicae Graph Theory*, 34(1):137–150, 2014.
- [IK98] Wilfried Imrich and Sandi Klavžar. A convexity lemma and expansion procedures for bipartite graphs. *European Journal of Combinatorics*, 19(6):677–685, 1998.
- [KM20] Kolja Knauer and Tilen Marc. On tope graphs of complexes of oriented matroids. *Discrete & Computational Geometry*, 63:377–417, 2020.
- [KS12] Sandi Klavžar and Sergey Shpectorov. Characterizing almost-median graphs ii. *Discrete Mathematics*, 312(2):462–464, 2012.
- [Mar18] Tilen Marc. *Cycling in hypercubes: doctoral thesis*. PhD thesis, Univerza v Ljubljani, Fakulteta za matematiko in fiziko, 2018.
- [MP14] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60:94–112, 2014.
- [Pol12] Norbert Polat. Weak geodesic topology and fixed finite subgraph theorems in infinite partial cubes II. fixed subgraph properties and infinite treelike partial cubes. *Discrete Mathematics*, 312(1):61–73, 2012.
- [XX25] Yan-Ting Xie and Shou-Jun Xu. Characterizing simplex graphs, 2025. arXiv:2503.17160, version 1.